

Chapter 10: Boolean Logic

1: Identify and use the standard symbols for logic gates

2: Define and understand the functions of the logic gates

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

By: Nouredine Tadjerout

Learning Objectives:

Candidates should be able to:

- 1 Identify and use the standard symbols for logic gates
- 2 Define and understand the functions of the logic gates
- 3 (a) Use logic gates to create given logic circuits from a:
 - (i) problem statement
 - (ii) logic expression
 - (iii) truth table

Notes and guidance

- See section 4 for logic gate symbols
- Including:
 - NOT
 - AND
 - OR
 - NAND
 - NOR
 - XOR (EOR)
 - the binary output produced from all the possible binary inputs
- NOT is a single input gate
- All other gates are limited to two inputs
- Circuits must be drawn for the statement given, without simplification
- Logic circuits will be limited to a maximum of three inputs and one output

Learning Objectives:

(b) Complete a truth table from a:

- (i) problem statement
- (ii) logic expression
- (iii) logic circuit

(c) Write a logic expression from a:

- (i) problem statement
- (ii) logic circuit
- (iii) truth table

- An example truth table with three inputs, for completion:

A	B	C	Output
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Resources:



<https://www.hoddereducation.co.uk/subjects/general/products/14-16/cambridge-igcse-and-o-level-computer-science-secon>

Resources:



Craig 'n' Dave For Students

CRAIG 'N' DAVE

To explore the entire series,
visit <https://craigndave.org/free-videos>

ICE Breaking: Recap previous chapter:

IGCSE Exam: Question 1

5 Parity checks are often used to check for errors that may occur during data transmission.

(a) A system uses **even parity**.

Tick (✓) to show whether the following three bytes have been transmitted correctly or incorrectly.

Received byte	Byte transmitted correctly	Byte transmitted incorrectly
1 1 0 0 1 0 0 0		
0 1 1 1 1 1 0 0		
0 1 1 0 1 0 0 1		

[3]

ICE Breaking: Recap previous chapter:

IGCSE Exam: Question 1

(b) A parity byte is used to identify which bit has been transmitted incorrectly in a block of data.

The word “F L O W C H A R T” was transmitted using nine bytes of data (one byte per character). A tenth byte, the parity byte, was also transmitted.

The following block of data shows all ten bytes received after transmission. The system uses even parity and column 1 is the parity bit.

	letter	column 1	column 2	column 3	column 4	column 5	column 6	column 7	column 8
byte 1	F	1	0	1	0	0	1	1	0
byte 2	L	1	0	1	0	1	1	0	0
byte 3	O	1	0	1	0	1	1	1	1
byte 4	W	1	0	1	1	0	1	1	1
byte 5	C	1	0	1	0	0	0	1	1
byte 6	H	0	0	1	0	1	0	0	0
byte 7	A	0	0	1	0	0	1	0	1
byte 8	R	1	0	1	1	0	0	1	0
byte 9	T	1	0	1	1	0	1	0	0
parity byte		1	0	1	1	1	1	1	0

(i) One of the bits has been transmitted incorrectly.

Write the byte number and column number of this bit:

Byte number

Column number

ICE Breaking: Recap previous chapter:

IGCSE Exam: Question 1

(ii) Explain how you arrived at your answer for part (b)(i).

.....

.....

.....

.....[2]

(c) Give the denary (base 10) value of the byte: **1 0 1 1 1 1 1 0**

.....

.....[1]

(d) A parity check may not identify that a bit has been transmitted incorrectly.

Describe **one** situation in which this could occur.

.....

.....

.....[1]

ICE Breaking: Recap previous chapter:

IGCSE Exam: Answer 1

5 (a) 1 mark per correctly placed tick

Received byte	Byte transmitted correctly	Byte transmitted incorrectly
1 1 0 0 1 0 0 0		✓
0 1 1 1 1 1 0 0		✓
0 1 1 0 1 0 0 1	✓	

[3]

(b) (i) byte number: 7

column number: 6

[2]

ICE Breaking: Recap previous chapter:

IGCSE Exam: Answer 1

(ii) Any **two** from:

- letter "A"(byte 7) transmitted as odd parity (three 1s)
- column 6 has odd parity (seven 1s)
- intersection of byte 7 and column 6 indicates incorrect bit value

[2]

(c) 190

[1]

(d) Any **one** from:

- 2 bits interchanged (e.g. $1 \rightarrow 0$ and $0 \rightarrow 1$) that won't change parity value
- even number of bits/digits are transposed
- If there are multiple errors in the same byte/column, that still produce the same parity bit, the error will not be detected

[1]

Introduction to logic gates and logic circuit

https://www.youtube.com/watch?v=0ClhfuXv_wg

The image is a YouTube video thumbnail with an orange background. In the top left, there is a black box with 'CAMBRIDGE' in white, and below it, another black box with 'IGCSE' in white. To the right of these, the text 'COMPUTER SCIENCE' is written in white, with '0478 | 0984' below it. In the center, the title 'INTRODUCTION TO LOGIC GATES & LOGIC CIRCUITS' is written in large, bold, black letters. To the right of the title, there are three logic gate symbols: an OR gate at the top, an AND gate in the middle, and a NOT gate at the bottom. In the top right corner, there are icons for a clock, a list, and a play button. In the bottom right corner, there is a black box with the text '6:06' in white. In the bottom left corner, the date '09/01/2023' is written in small black text, and in the bottom center, the text 'By: Nouredine Tadjerout' is written in small black text.

CAMBRIDGE

IGCSE

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10

**INTRODUCTION TO LOGIC GATES
& LOGIC CIRCUITS**

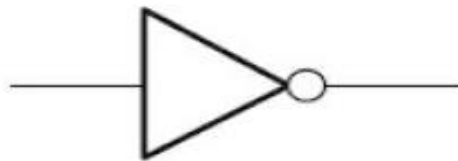
09/01/2023

By: Nouredine Tadjerout

6:06

1: Identify and use the standard symbols for logic gates

Six different logic gates will be considered in this chapter:



NOT gate



AND gate



OR gate



NAND gate



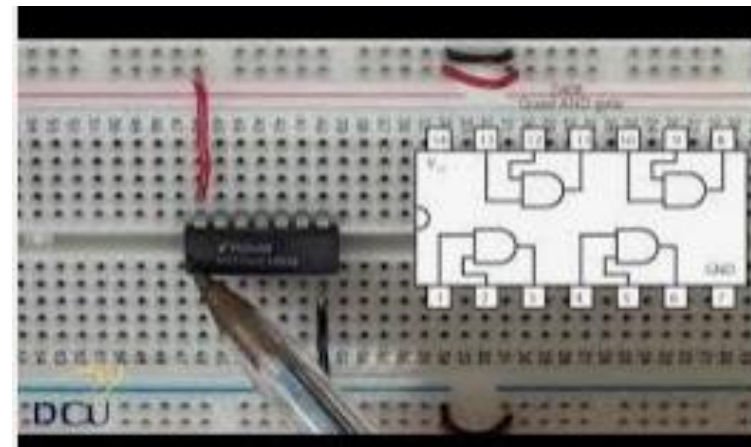
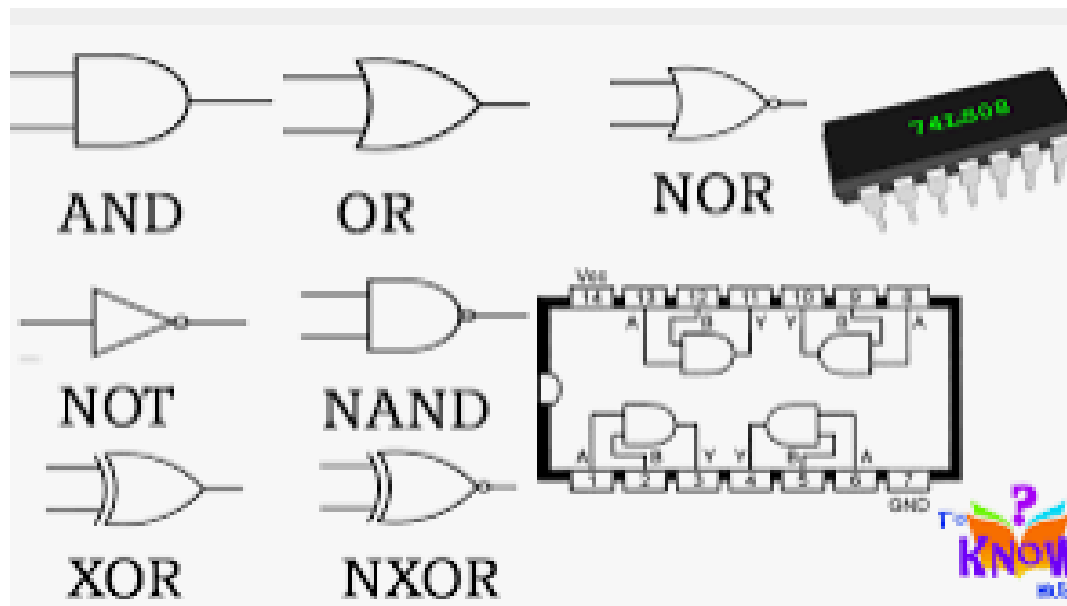
NOR gate



XOR gate

1: Identify and use the standard symbols for logic gates

Electronic circuits in computers, many new memories and controlling devices are made up of thousands of **LOGIC GATES**. Logic gates take binary inputs and produce a binary output also electronic circuits have to make decisions. They look at two or more inputs and use these to determine the outputs from the circuit and the process of doing this uses electronic logic, which is based on digital switches called gates. When logic gates are combined together to carry out a particular function, such as controlling a robot, they form a logic circuit.



1: Identify and use the standard symbols for logic gates

Truth tables: Truth tables are used to trace the output from a logic gate or logic circuit.

When constructing truth tables, all possible combinations of 1s and 0s which can be input are considered.

For the NOT gate (one input) there are only 2^1 (2) possible binary combinations. For all other gates (two inputs), there are 2^2 (4) possible binary combinations.

For logic circuits, the number of inputs can be more than 2; for example: three inputs give a possible 2^3 (8) binary combinations and four inputs, the number of possible binary combinations is 2^4 (16).

1: Identify and use the standard symbols for logic gates

Truth tables:

Truth tables for two, three and four inputs

2^2 (4)

A	B
0	0
0	1
1	0
1	1

2^3 (8)

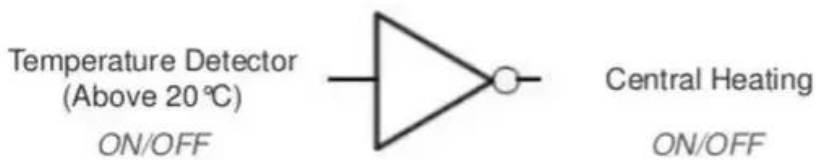
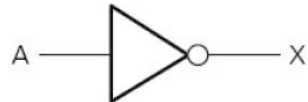
A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

2^4 (16)

A	B	C	D
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

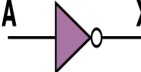
2: Define and understand the functions of the logic gates

NOT gate:



If the temperature is above 20°C then the Central Heating is switched off.

If the temperature is below 20°C then the Central Heating is switched on.

Boolean Expression	Logic Diagram Symbol	Truth Table						
$X = A'$		<table><tr><th>A</th><th>X</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	A	X	0	1	1	0
A	X							
0	1							
1	0							

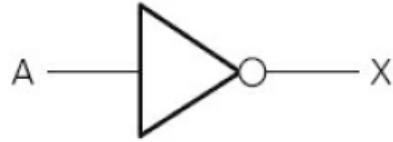
When the temperature falls below 20c the **Not** gate will set on the central heating system (cool huh)

How to write this:

$X = \text{NOT } A$ (logic notation)
 $X = \bar{a}$ (Boolean algebra)

Description:	Truth table:	How to write this:								
The output, X, is 1 if: the input, A, is 0 By: Nouredine Tadjerout	▼ Table 10.2 <table><tr><th>Input</th><th>Output</th></tr><tr><th>A</th><th>X</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	Input	Output	A	X	0	1	1	0	$X = \text{NOT } A$ (logic notation) $X = \overline{A}$ (Boolean algebra)
Input	Output									
A	X									
0	1									
1	0									

1: Identify and use the standard symbols for logic gates



Click the link below to see an example NOT gate:

<https://www.wisc-online.com/learn/technical/electronics-dc/dce16114/basic-logic-gates>

2: Define and understand the functions of the logic gates

AND gate:



How to write this:

X = A AND B (logic notation)

X = a · b (Boolean algebra)

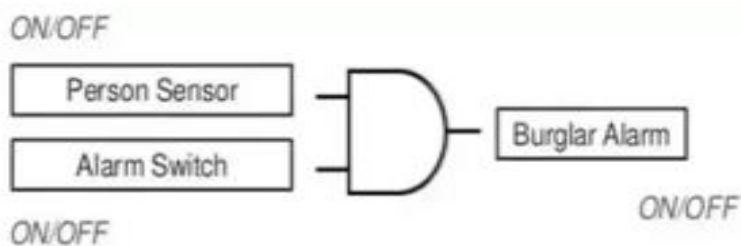
Boolean Expression

$X = A \cdot B$

Logic Diagram Symbol

Truth Table

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1



Description:	Truth table:	How to write this:
The output, X, is 1 if: both inputs, A and B, are 1	▼ Table 10.3	

If both the Person Sensor AND the Alarm Switch are on then the Burglar Alarm is activated.



(Note: the three symbols in the Boolean algebra have the following meaning:
· represents the AND operation
+ represents the OR operation
a bar above the letter, e.g. $\bar{a}$, represents the NOT operation.)

1: Identify and use the standard symbols for logic gates



Click the link below to see an example of AND gate

<https://www.wisc-online.com/learn/technical/electronics-dc/dce16114/basic-logic-gates>

2: Define and understand the functions of the logic gates

OR gate:



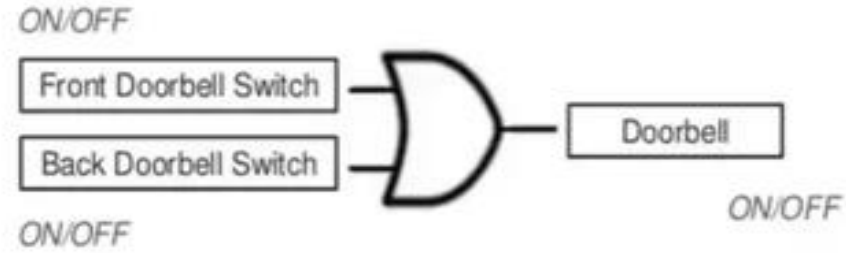
Boolean Expression

$X = A + B$

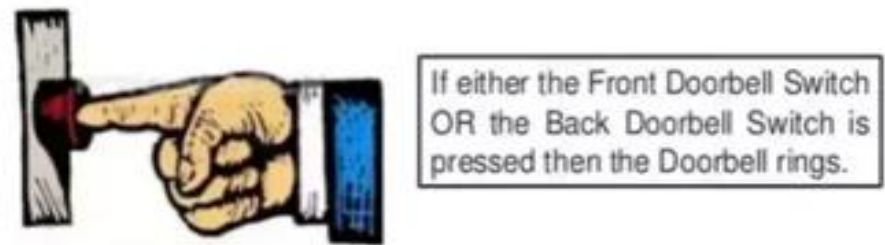
Logic Diagram Symbol

Truth Table

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1



Description:	Truth table:	How to write this:
The output, X, is 1 if: either input, A or B, or both, are 1	▼ Table 10.4	



How to write this:

X = A OR B (logic notation)

X = a + b (Boolean algebra)

(Note: the three symbols in the Boolean algebra have the following meaning:
• represents the AND operation
+ represents the OR operation
a bar above the letter, e.g. $\bar{a}$, represents the NOT operation.)

1: Identify and use the standard symbols for logic gates



Click the link below to see an example of OR gate:

<https://www.wisc-online.com/learn/technical/electronics-dc/dce16114/basic-logic-gates>

2: Define and understand the functions of the logic gates

NAND gate (NOT AND):



How to write this:

$X = A \text{ NAND } B$ (logic notation)

$X = \overline{a \cdot b}$ (Boolean algebra)

Description:	Truth table:	How to write this:																		
The output, X, is 1 if: input A AND input B are NOT both 1	▼ Table 10.5																			
	<table><tr><th colspan="2">Inputs</th><th>Output</th></tr><tr><th>A</th><th>B</th><th>X</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>		Inputs		Output	A	B	X	0	0	1	0	1	1	1	0	1	1	1	0
	Inputs		Output																	
	A	B	X																	
	0	0	1																	
	0	1	1																	
	1	0	1																	
1	1	0																		
X = A NAND B (logic notation)																				
$X = \overline{A \cdot B}$ (Boolean algebra)																				

2: Define and understand the functions of the logic gates

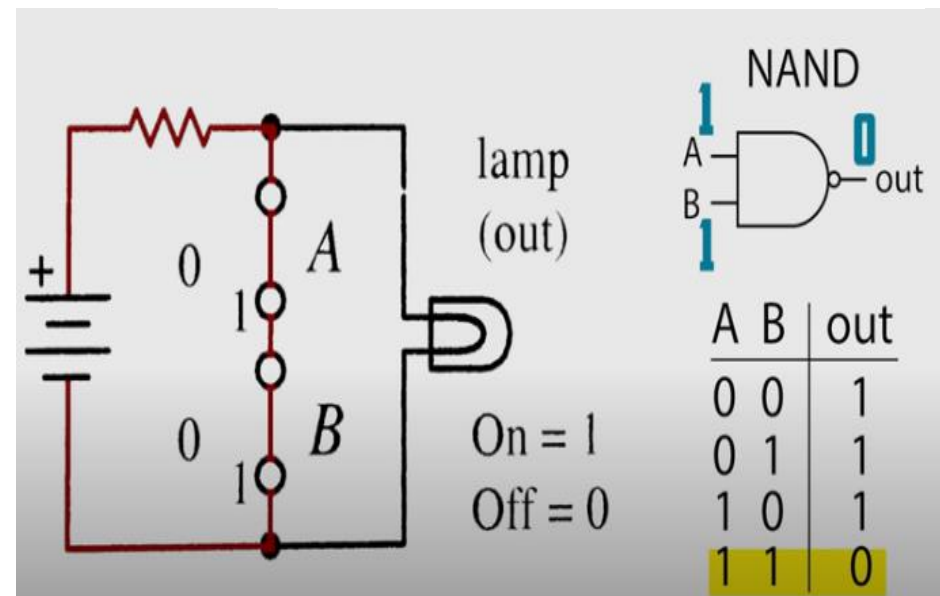
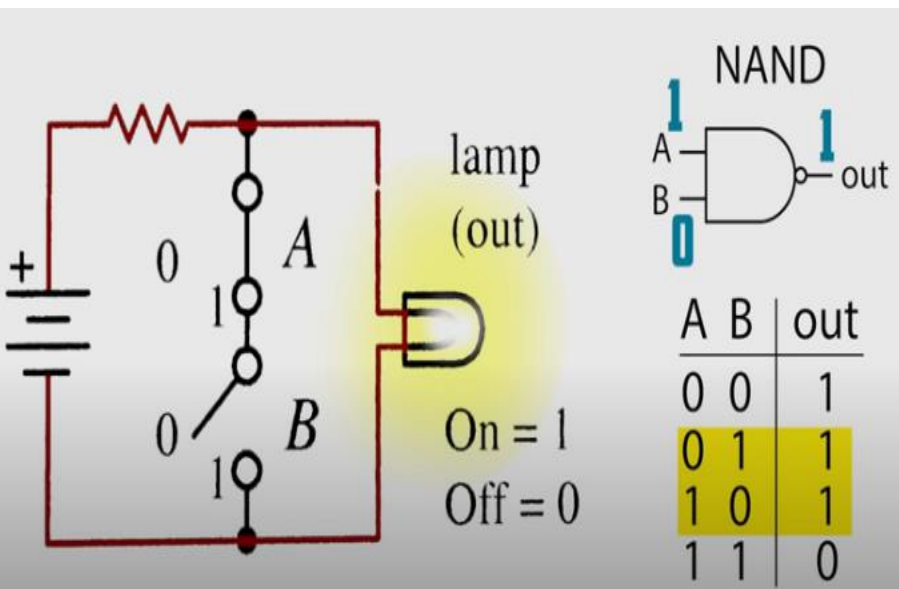
NAND gate (NOT AND):



How to write this:

$X = A \text{ NAND } B$ (logic notation)

$X = \overline{a \cdot b}$ (Boolean algebra)



1: Identify and use the standard symbols for logic gates



NAND gate is just the opposite of **AND** gate.

Click the link below to see an example of an NAND:

<https://www.wisc-online.com/learn/technical/electronics-dc/dce16114/basic-logic-gates>

2: Define and understand the functions of the logic gates

NOR gate (NOT OR):



How to write this:

$X = A \text{ NOR } B$ (logic notation)

$X = \overline{a + b}$ (Boolean algebra)

Description:	Truth table:	How to write this:
The output, X, is 1 if: neither input A nor input B is 1	▼ Table 10.6	
		X = A NOR B (logic notation)
		X = $\overline{A + B}$ (Boolean algebra)

2: Define and understand the functions of the logic gates

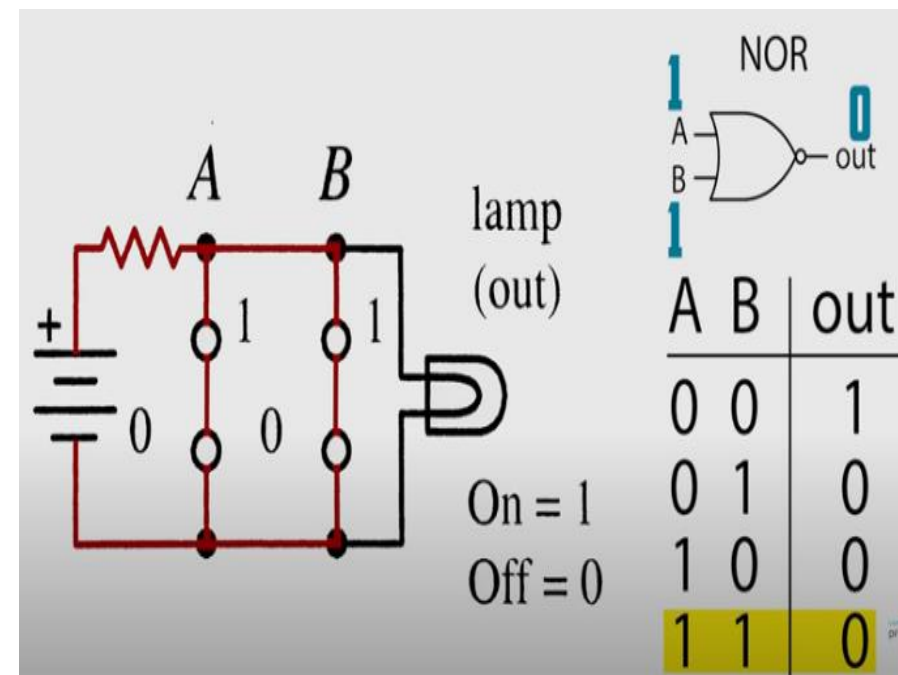
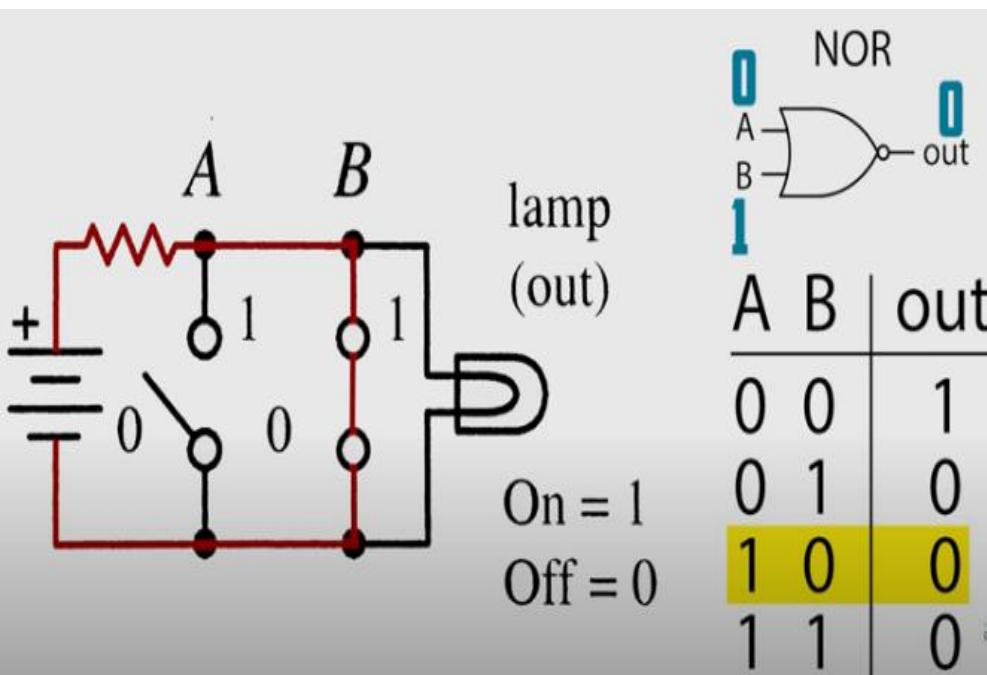
NOR gate (NOT OR):



How to write this:

$X = A \text{ NOR } B$ (logic notation)

$X = \overline{a + b}$ (Boolean algebra)



1: Identify and use the standard symbols for logic gates



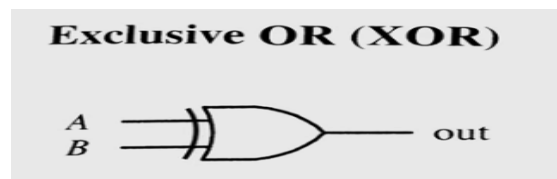
NOR gate is just the opposite of **OR** gate.

Click the link below to see an example of NOR gate.

<https://www.wisc-online.com/learn/technical/electronics-dc/dce16114/basic-logic-gates>

2: Define and understand the functions of the logic gates

XOR (Exclusive OR) gate:



If both states are ON or both state are OFF , we are not going to do anything because the output will be OFF, However If both states are not the same then do something



Description:

The output, X, is 1 if:
(input A is 1 AND input B is 0) OR
(input A is 0 AND input B is 1)

Truth table:

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

How to write this:

$X = A \text{ XOR } B$ (logic notation)

$X = (a \cdot \bar{b}) + (\bar{a} \cdot b)$ (Boolean algebra)

NOTE: this is sometimes written as:

$(a + b) \cdot (\bar{a} \cdot \bar{b})$

(Note: the three symbols in the Boolean algebra have the following meaning:

• represents the AND operation

+ represents the OR operation

a bar above the letter, e.g. $\bar{a}$, represents the NOT operation.)

09/01/2023

By: Nouredine Tadjerout

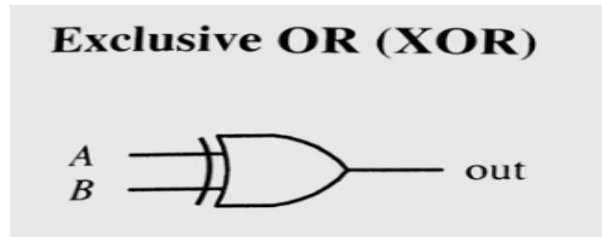
2: Define and understand the functions of the logic gates

XOR (Exclusive OR) gate:



Description:	Truth table:	How to write this:																		
The output, X, is 1 if: (input A is 1 AND input B is 0) or (input A is 0 AND input B is 1)	▼ Table 10.7 <table> <tr> <th colspan="2">Inputs</th><th>Output</th></tr> <tr> <th>A</th><th>B</th><th>X</th></tr> <tr> <td>0</td><td>0</td><td>0</td></tr> <tr> <td>0</td><td>1</td><td>1</td></tr> <tr> <td>1</td><td>0</td><td>1</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </table>	Inputs		Output	A	B	X	0	0	0	0	1	1	1	0	1	1	1	0	$X = A \text{ XOR } B$ (logic notation) $X = (A \cdot \bar{B}) + (\bar{A} \cdot B)$ (Boolean algebra) NOTE: this is sometimes written as: $(A + B) \cdot \overline{(A \cdot B)}$
Inputs		Output																		
A	B	X																		
0	0	0																		
0	1	1																		
1	0	1																		
1	1	0																		

1: Identify and use the standard symbols for logic gates



XOR (Exclusive OR) gate:

If both states are ON or both state are OFF , we are not going to do anything because the output will be OFF, However If both states are not the same then do something

Click the link below to see an example of light stairway.

<https://www.wisc-online.com/learn/technical/electronics-dc/dce16114/basic-logic-gates>

2: Define and understand the functions of the logic gates

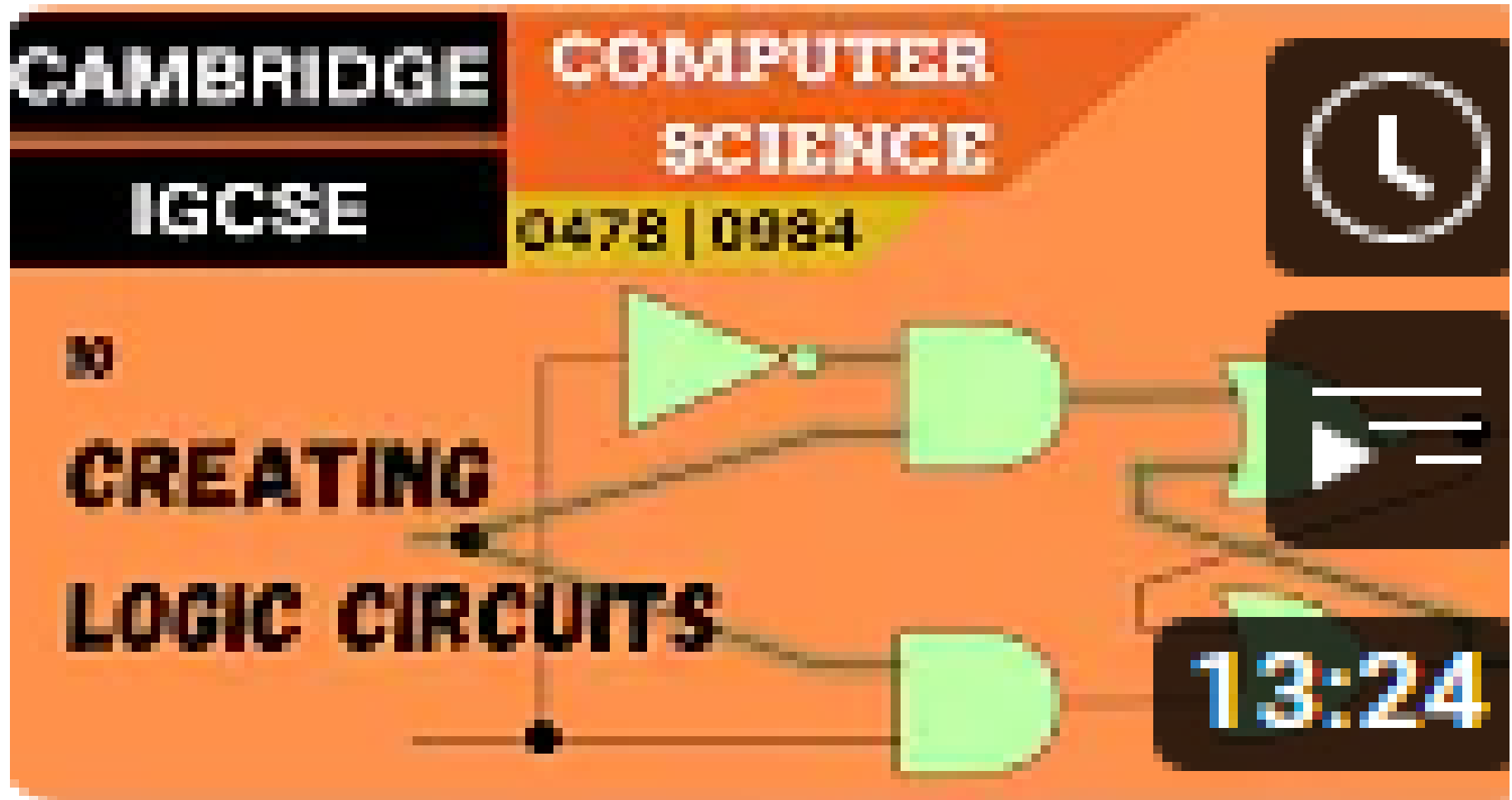
AND	OR	NOT																																				
																																						
<table><tr><th>A</th><th>B</th><th>Q</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	Q	0	0	0	0	1	0	1	0	0	1	1	1	<table><tr><th>A</th><th>B</th><th>Q</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	Q	0	0	0	0	1	1	1	0	1	1	1	1	<table><tr><th>A</th><th>Q</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	A	Q	0	1	1	0
A	B	Q																																				
0	0	0																																				
0	1	0																																				
1	0	0																																				
1	1	1																																				
A	B	Q																																				
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1	0	1																																				
1	1	1																																				
A	Q																																					
0	1																																					
1	0																																					

$\bar{A}$ which is also written as NOT A
 $A.B$ which is also written as A AND B
 $A + B$ which is also written as A OR B

NAND	NOR	XOR																																													
																																															
<table><tr><th>A</th><th>B</th><th>Q</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Q	0	0	1	0	1	1	1	0	1	1	1	0	<table><tr><th>A</th><th>B</th><th>Q</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Q	0	0	1	0	1	0	1	0	0	1	1	0	<table><tr><th>A</th><th>B</th><th>Q</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Q	0	0	0	0	1	1	1	0	1	1	1	0
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A	B	Q																																													
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Logic circuits, truth tables and logic expressions

<https://www.youtube.com/watch?v=Y7VZ3XT-QUs>



Logic circuits, truth tables and logic expressions

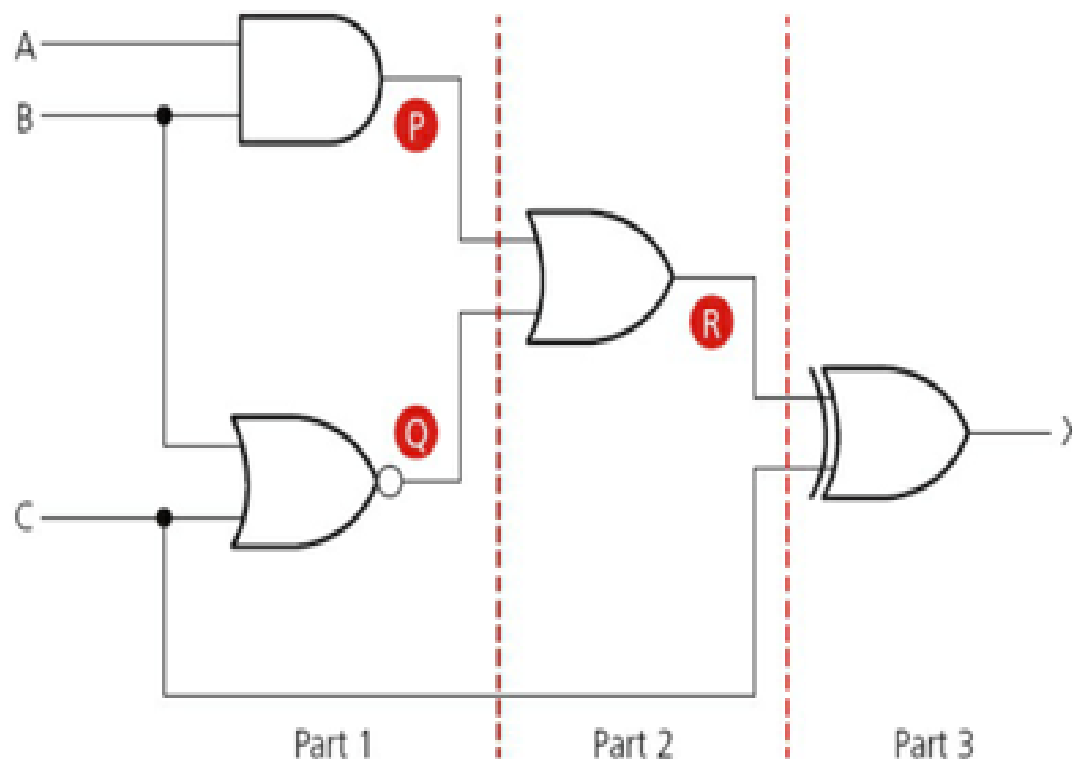
<https://youtu.be/UdirtczcTeQ>

**Creating a
truth table
from a logic
circuit**

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 1:

Produce a truth table for the following logic circuit (note the use of • at junctions):

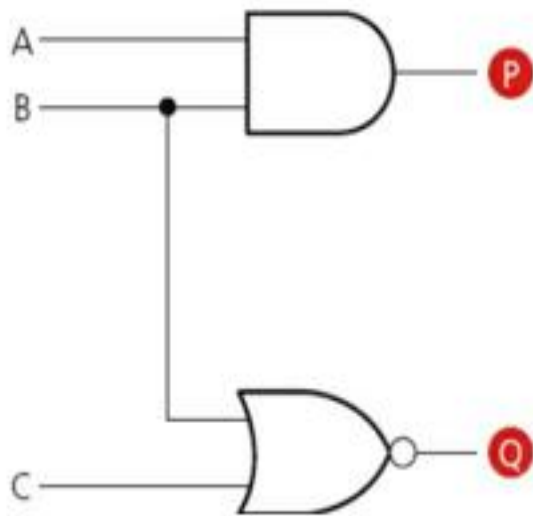


There are three inputs to this logic circuit (AND, OR, XOR gate), therefore there will be eight possible binary values which can be input. To show step-wise how the truth table is produced, the logic circuit has been split up into three parts and intermediate values are shown as P, Q and R.

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 1:

The value of P is found from the **AND** gate where the inputs are **A** and **B**. The value of Q is found from the **NOR** gate where the inputs are B and C. An intermediate truth table is produced using the logic function descriptions in **slide 6**

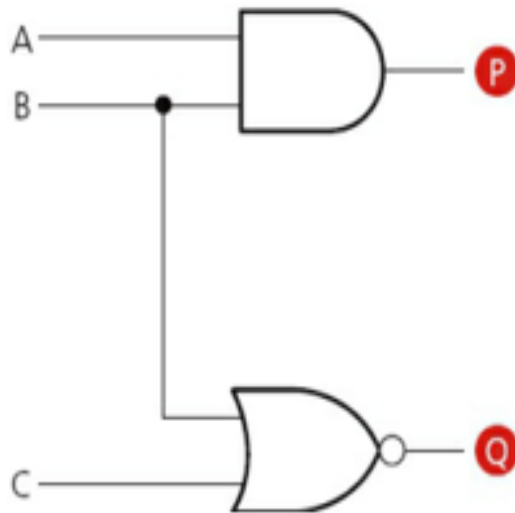


. represents the AND operation
+ represents the OR operation

Input values			Output values	
A	B	C	P	Q
0	0	0	0	1
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	1
1	0	1	0	0
1	1	0	1	0
1	1	1	1	0

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 1:



. represents the AND operation
+ represents the OR operation

The value of P is found from the **AND** gate where the inputs are **A** and **B**. The value of Q is found from the **NOR** gate where the inputs are B and C. An intermediate truth table is produced using the logic function descriptions in **slide 6**

Input values			Output values	
A	B	C	P	Q
0	0	0	0	1
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	1
1	0	1	0	0
1	1	0	1	0
1	1	1	1	0

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 1:

The final part of the logic circuit has R and C as inputs and the final output, X:



This gives the third intermediate truth table:

Inputs		Output
R	C	X
1	0	1
0	1	1
0	0	0
0	1	1
1	0	1
0	1	1
1	0	1
1	1	0

. represents the AND operation
+ represents the OR operation

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 1:

The intermediate values can be left out of the final truth table, but it is good practice to leave them in until you become confident about producing the truth tables. The final truth table would then look like this:

Input values			Output
A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 1:

Putting all three intermediate truth tables together produces the final truth table which represents the original logic circuit:

Input values			Intermediate values			Output
A	B	C	P	Q	R	X
0	0	0	0	1	1	1
0	0	1	0	0	0	1
0	1	0	0	0	0	0
0	1	1	0	0	0	1
1	0	0	0	1	1	1
1	0	1	0	0	0	1
1	1	0	1	0	1	1
1	1	1	1	0	1	0

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

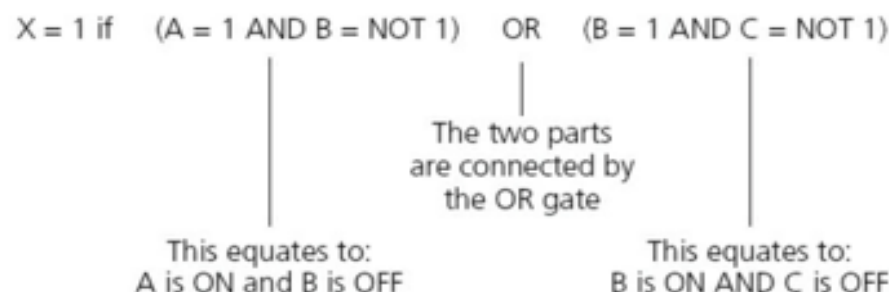
Example 2:

A safety system uses three inputs to a logic circuit. An **alarm, X, sounds** if input **A represents ON** **and** input **B represents OFF**; **or** if input **B represents ON** **and** input **C represents OFF**.

Produce a logic circuit and truth table to show the conditions which cause the output **X** to be **1**.

The first thing to do is to write down the **logic statement** representing the scenario in this example. To do this, it is necessary to recall that **ON = 1** and **OFF = 0** and also that **0** is usually considered to be **NOT 1**.

So we get the following logic statement:



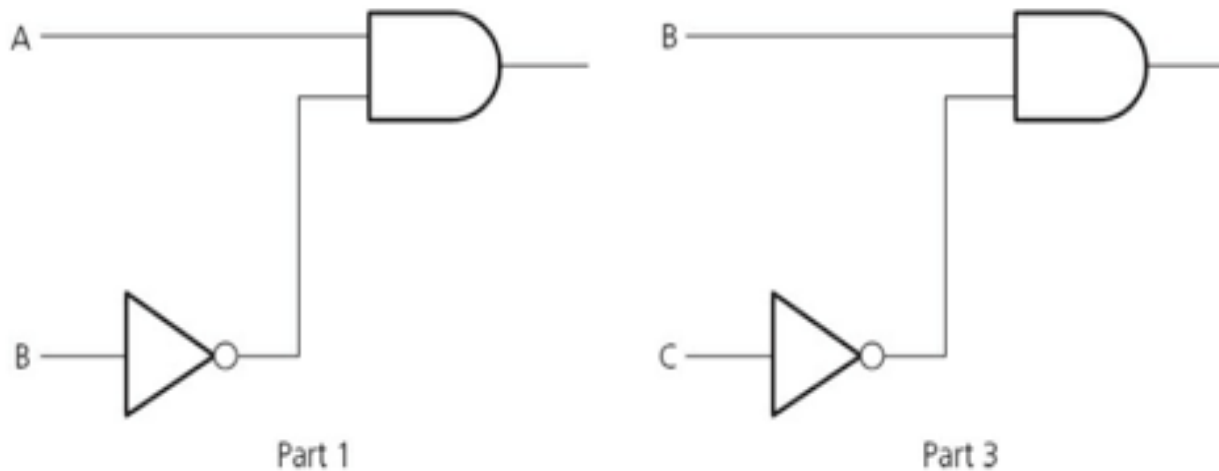
Note: this statement can also be written in Boolean algebra as:
 $(a \cdot \bar{b}) + (b \cdot \bar{c})$

. represents the AND operation
+ represents the OR operation

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 2:

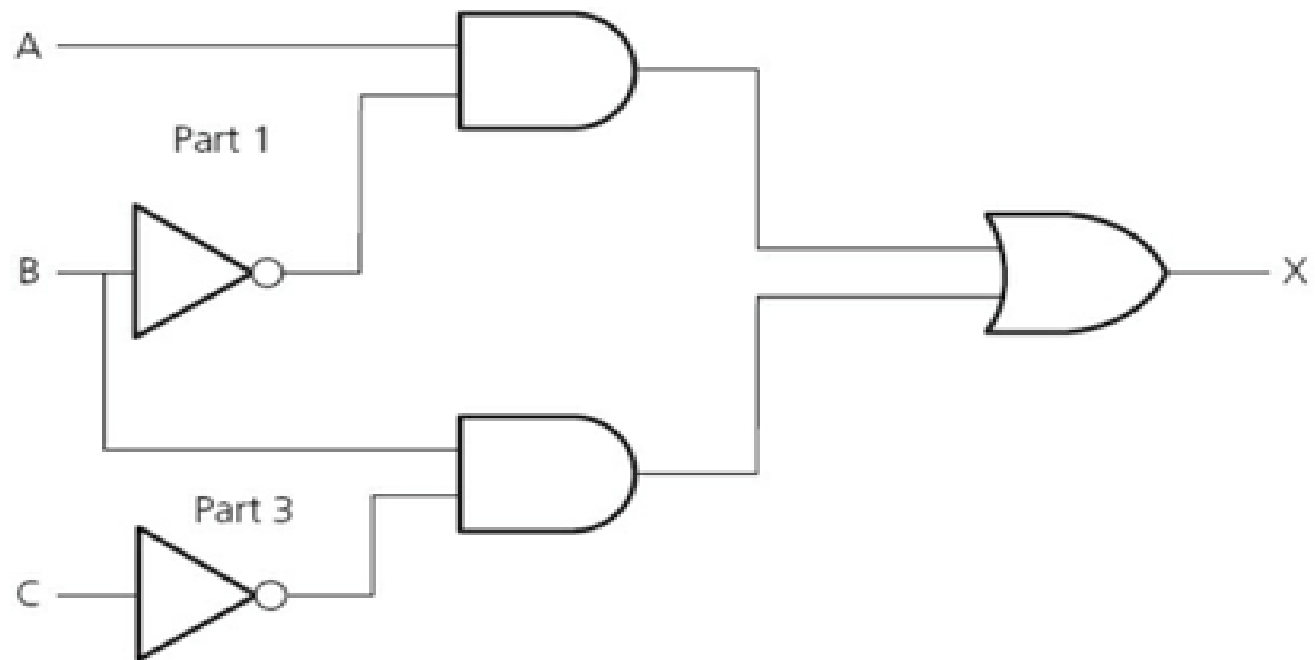
The logic circuit is made up of three parts as shown in the logic statement. We will produce the logic gate for the first part and the third part. Then join both parts together with the **OR** gate.



3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 2:

Now combining both parts with the OR gate gives us:



3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 2:

Produce the truth table using the original logic statement:

Inputs			Intermediate values		Output
A	B	C	(A=1 AND B=NOT 1)	(B=1 AND C=NOT 1)	X
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	0	1
1	1	0	0	1	1
1	1	1	0	0	0

(Note: it is optional to leave in the intermediate values or to remove them giving a four-column truth table with headings: A, B, C, X.)

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Let's look at example 1 :

● Sample question

2 Show why $(a \cdot \bar{b}) + (\bar{a} \cdot b)$ and $(a + b) \cdot \overline{(a \cdot b)}$ both represent an XOR gate.

Answer:

Use of a truth table:

Inputs							
a	b	$(a \cdot \bar{b})$	$(\bar{a} \cdot b)$	$(a \cdot \bar{b}) + (\bar{a} \cdot b)$	$(a + b)$	$\overline{(a \cdot b)}$	$(a + b) \cdot \overline{(a \cdot b)}$
0	0	0	0	0	0	1	0
0	1	0	1	1	1	1	1
1	0	1	0	1	1	1	1
1	1	0	0	0	1	0	0

Columns 5 and 8 are the same which proves that the two expressions are the same and the output values correspond with an XOR gate.

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Let's look at example 2:

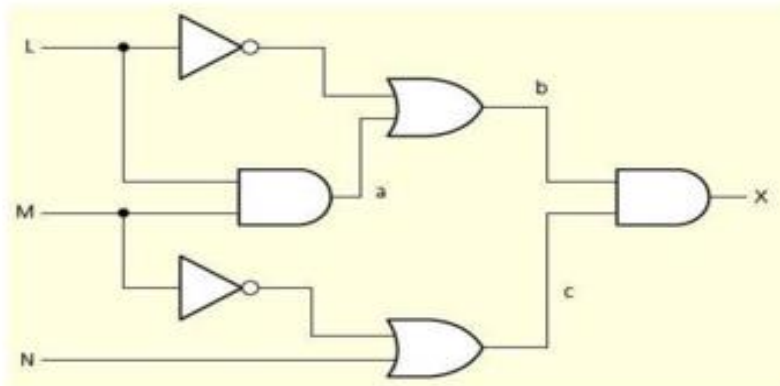
● Sample question

3 a Draw the logic circuit for the following logic statement:

$$(L \cdot M + \bar{L}) \cdot (\bar{M} + N)$$

b Complete the truth table for the above logic statement.

Answer:



Inputs			Working			Output
L	M	N	a	b	c	X
0	0	0	0	1	1	1
0	0	1	0	1	1	1
0	1	0	0	1	0	0
0	1	1	0	1	1	1
1	0	0	0	0	1	0
1	0	1	0	0	1	0
1	1	0	1	1	0	0
1	1	1	1	1	1	1

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 3:

A wind turbine has a safety system which uses **three inputs** to a logic circuit. A certain combination of conditions results in an **output, X**, from the logic circuit being **equal to 1**. When the value of **X = 1** then the wind turbine is **shut down**.

The following table shows which parameters are being monitored and form the three inputs to the logic circuit.

Parameter description	Parameter	Binary value	Description of condition
turbine speed	S	0	≤ 1000 rpm
		1	> 1000 rpm
bearing temperature	T	0	$\leq 80^{\circ}\text{C}$
		1	$> 80^{\circ}\text{C}$
wind velocity	W	0	≤ 120 kph
		1	> 120 kph

bearing.



Revolutions per minute (abbreviated **rpm**, **RPM**, is the number of turns in one minute. It is a unit of rotational speed or the frequency of rotation around a fixed axis.

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 3:

The **output, X**, will have a value of **1** if any of the following combination of conditions occur:

- **either** turbine speed ≤ 1000 rpm **and** bearing temperature $> 80^{\circ}\text{C}$
- **or** turbine speed > 1000 rpm **and** wind velocity > 120 kph
- **or** bearing temperature $\leq 80^{\circ}\text{C}$ **and** wind velocity > 120 kph.

Design the logic circuit and complete the truth table to produce a value of **X = 1** when any of the three conditions above occur.

This time a real situation is given and it is necessary to convert the information into a logic statement and then produce the logic circuit and truth table.

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 3:

Stage 1

The first thing to do is to convert each of the three statements into logic statements. Use the information given in the table and the **three** condition statements to find how the three parameters, **S**, **T** and **W**, are linked. We usually look for the key words **AND**, **OR** and **NOT** when converting actual statements into logic.

We end up with the following three logic statements:

a): turbine speed ≤ 1000 rpm **and** bearing temperature $> 80^{\circ}\text{C}$

logic statement: (**S = NOT 1 AND T = 1**)

b): turbine speed > 1000 rpm **and** wind velocity > 120 kph

logic statement: (**S = 1 AND W = 1**)

c): bearing temperature $\leq 80^{\circ}\text{C}$ **and** wind velocity > 120 kph

logic statement: (**T = NOT 1 AND W = 1**)

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 3:

Stage 2: This now produces three intermediate logic circuits:

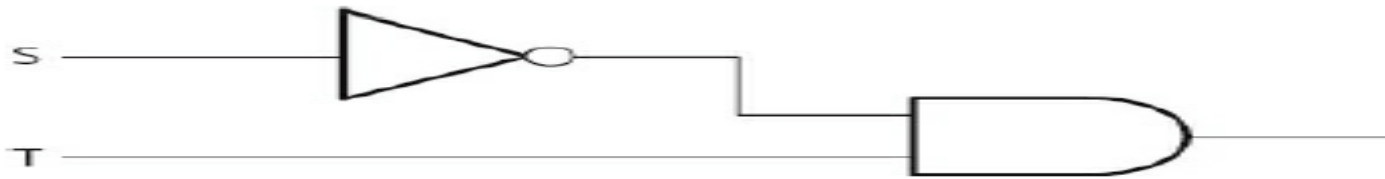


Figure 3.20

ii



Figure 3.21

iii

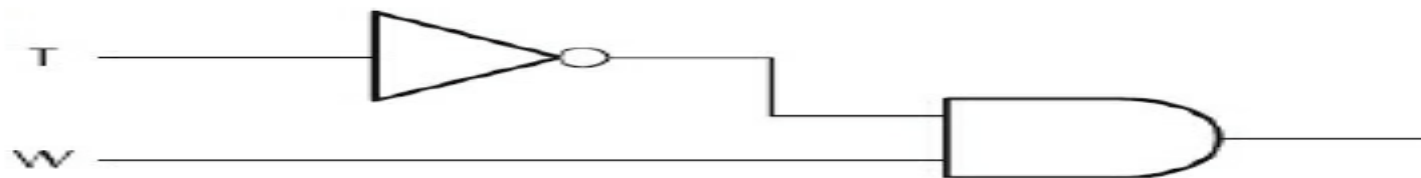


Figure 3.22

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 3:

Stage 2: Each of the three original statements were joined together by the word **OR**. Thus we need to join all of the **three intermediate logic** circuits by **two OR** gates to get the final logic circuit. We will start by joining (i) **and** (ii) together using an **OR** gate

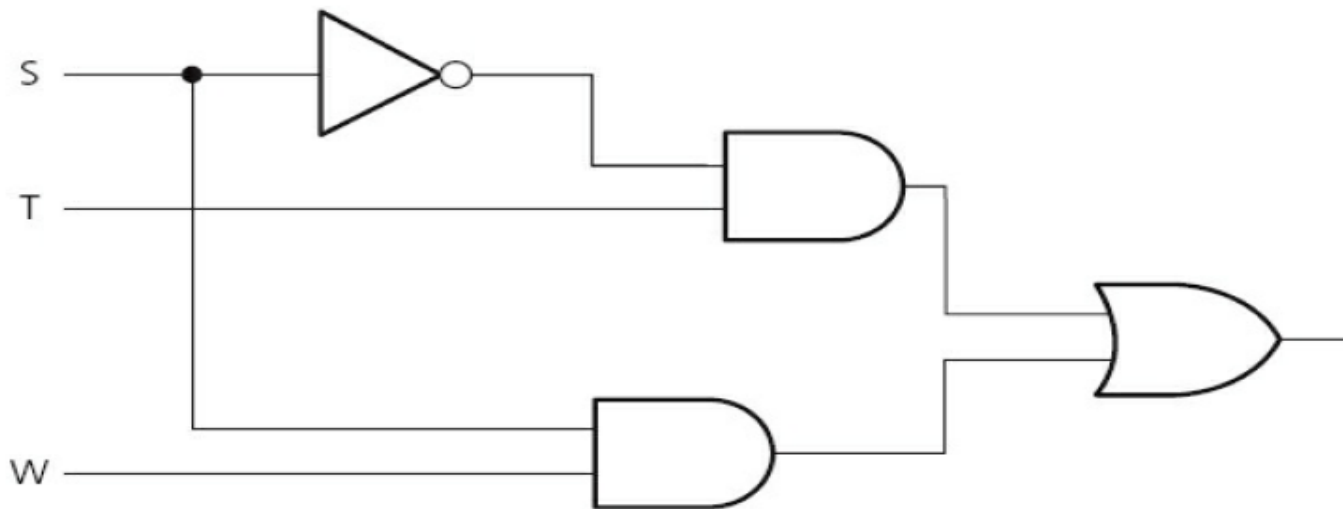


Figure 3.23

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 3:

Stage 2: Finally, we connect the logic circuit in **Figure 3.23** to **Figure 3.22** to obtain the answer:

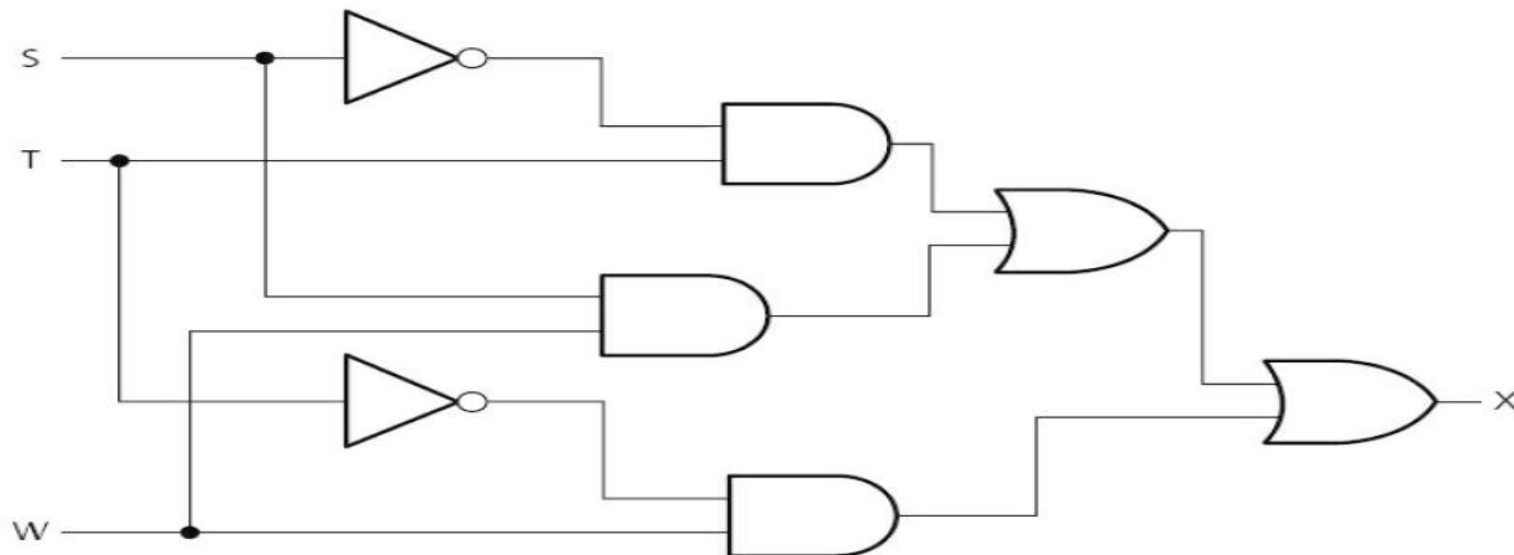


Figure 3.24

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Example 3:

Stage 2:

The final part is to produce the truth table.

There were three parts to the problem, so the truth table will first evaluate each part.

Then, by applying OR gates, as shown below, the final value, X, is obtained:

a): (S = NOT 1 AND T = 1)

b): (S = 1 AND W = 1)

c): (T = NOT 1 AND W = 1)

We find the outputs from parts (i) and (ii) and then OR these two outputs together to obtain a new intermediate, which we will label part (iv). We then OR parts (iii) and (iv) together to get the value of X.

Inputs			Intermediate values				Output
S	T	W	(i) (S=NOT 1 AND T=1)	(ii) (S=1 AND W=1)	(iii) (T=NOT 1 AND W=1)	(iv)	X
0	0	0	0	0	0	0	0
0	0	1	0	0	1	0	1
0	1	0	1	0	0	1	1
0	1	1	1	0	0	1	1
1	0	0	0	0	0	0	0
1	0	1	0	1	1	1	1
1	1	0	0	0	0	0	0
1	1	1	0	1	0	1	1

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Logic circuits in the real world

Practical Activity:

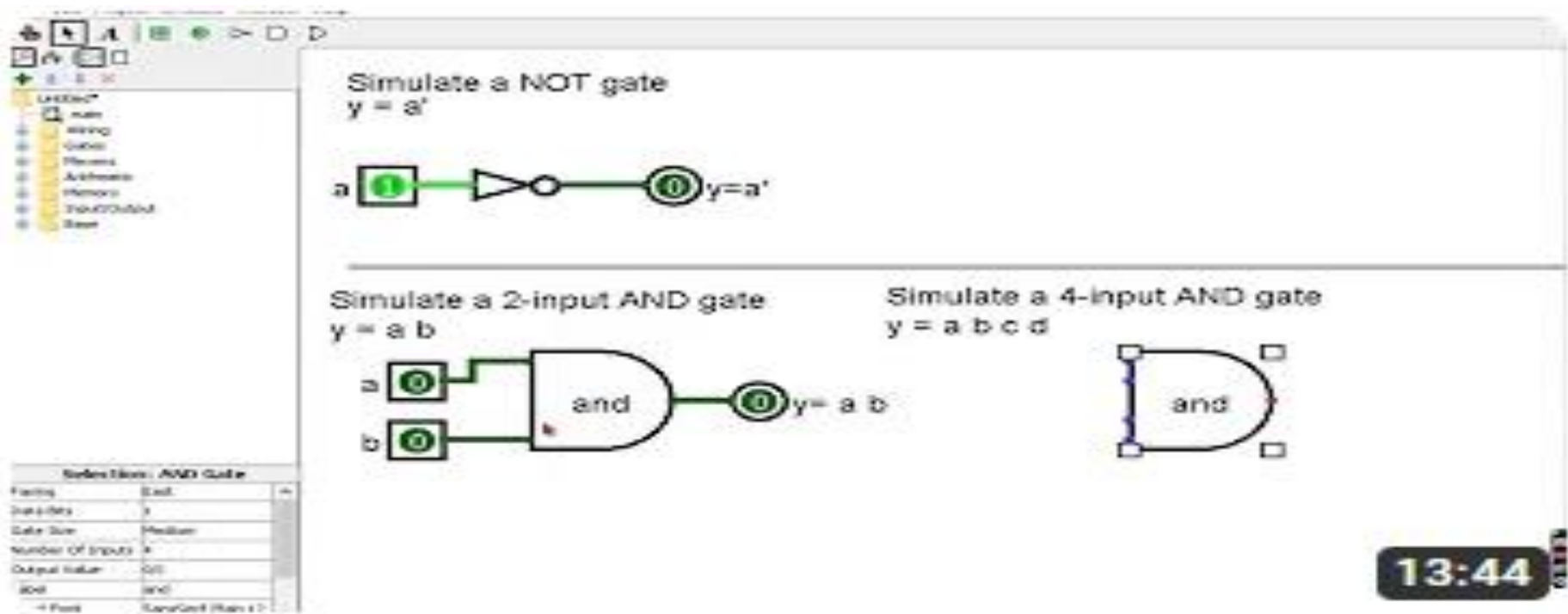
After you watched the video then click on the link below to download the **Logisim** for free. An educational tool for designing and simulating digital logic circuits, featuring a simple-to-learn interface, hierarchical circuits.

<https://sourceforge.net/projects/circuit/files/2.7.x/2.7.1/logisim-win-2.7.1.exe/download>

Practical Activity:

Introduction to Logisim - Simulating the Basic Logic Gates (NOT, AND, OR)

<https://www.youtube.com/watch?v=FXzim08Gjwk>



3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Logic circuits in the real world

Electronics companies need to consider the cost of components, fabrication and time, when designing and building logic circuits. Two possible ways for electronics companies can review logic circuit design:

First method is to use 'off-the-shelf' logic units and build up the logic circuit as a number of 'building blocks'.

Second method involves simplifying the logic circuit as far as possible; this may be necessary where room is at a premium (e.g. in building circuit boards for use in satellites to allow space exploration).

3: Use logic gates to create given logic circuits and Truth Table and write a logic expression

Logic circuits in the real world

Using logic 'building blocks'

One common 'building block' is the NAND gate. It is possible to build up any logic gate, and therefore any logic circuit, by simply linking together a number of NAND gates, such as:

- the AND gate



Figure 3.29 AND gate made from NAND gates

- the OR gate

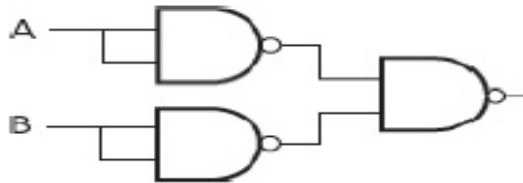


Figure 3.30 OR gate made from NAND gates

- the NOT gate

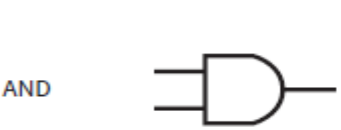


Figure 3.31 NOT gate made from NAND gates

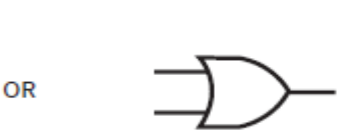
Summary of Logic gates and logic circuits



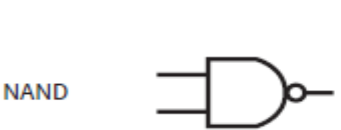
A	X
0	1
1	0



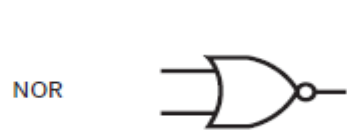
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1



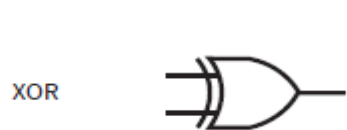
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1



A	B	X
0	0	1
0	1	1
1	0	1
1	1	0



A	B	X
0	0	1
0	1	0
1	0	0
1	1	0



A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

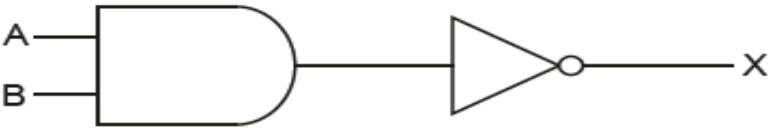
Summary

- A logic scenario can be described by a problem statement or a logic expression.
- A logic expression comprises logic propositions and Boolean operators.
- Logic circuits are constructed from logic gates.
- The operation of a logic gate matches that of a Boolean operator.
- The outcome of a logic expression or a logic circuit can be expressed as a truth table.
- A logic expression can be created from a truth table using the rows that provide a 1 output.

IGCSE Past Exam: Question 1

4 (a) Identify the name **and** draw the **single** logic gate that can replace the given logic circuits.

(i)



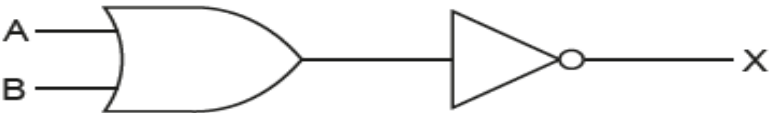
Name of gate:

Drawing of gate:



[2]

(ii)



Name of gate:

Drawing of gate:



[2]

IGCSE Past Exam: Question 1

(b) Complete the truth table for the given logic statement:

$$X = (((A \text{ OR } C) \text{ AND } (\text{NOT } A \text{ AND } \text{NOT } C)) \text{ XOR } B)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

IGCSE Past Exam: Answer 1

Question	Answer	Marks
4(a)(i)	– NAND 	2
4(a)(ii)	– NOR 	2

Question	Answer					Marks
4(b)	A	B	C	Working space	X	4
	0	0	0		0	
	0	0	1		0	
	0	1	0		1	
	0	1	1		1	
	1	0	0		0	
	1	0	1		0	
	1	1	0		1	
	1	1	1		1	
	4 marks for 8 correct outputs 3 marks for 6 or 7 correct outputs 2 marks for 4 or 5 correct outputs 1 mark for 2 or 3 correct outputs					

IGCSE Past Exam: Question 2

2 Rajesh creates a logic circuit.

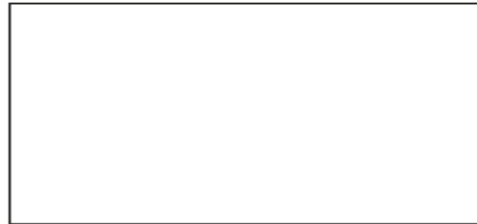
He uses three different logic gates in his circuit. Each logic gate has a maximum of **two** inputs.

He describes the logic of each gate.

(a) “The only time the output will be 1 is when both inputs are 1.”

State the single logic gate

Draw the single logic gate:



[2]

(b) “The only time the output will be 1 is when both inputs are 0.”

State the single logic gate

Draw the single logic gate:



[2]

IGCSE Past Exam: Question 2

(c) “The only time the output will be 0 is when both inputs are 1.”

State the single logic gate

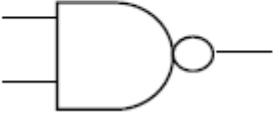
Draw the single logic gate:



[2]

IGCSE Past Exam: Answer 2

Question	Answer	Marks
2(a)	1 mark for correct name, 1 mark for correct gate symbol - AND 	2
2(b)	1 mark for correct name, 1 mark for correct gate symbol - NOR 	2

Question	Answer	Marks
2(c)	1 mark for correct name, 1 mark for correct gate symbol - NAND 	2

IGCSE Past Exam: Question 3

6 Consider the given logic statement:

$$X = (((A \text{ XOR } B) \text{ AND } C) \text{ OR NOT } C)$$

(a) Draw a logic circuit to match the given logic statement.

All logic gates must have a maximum of **two** inputs. Do **not** attempt to simplify the logic statement.



IGCSE Past Exam: Question 3

(b) Complete the truth table for the given logic statement.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

IGCSE Past Exam: Answer 3

Question	Answer	Marks
6(a)	One mark for each correct gate.	4

IGCSE Past Exam: Answer 3

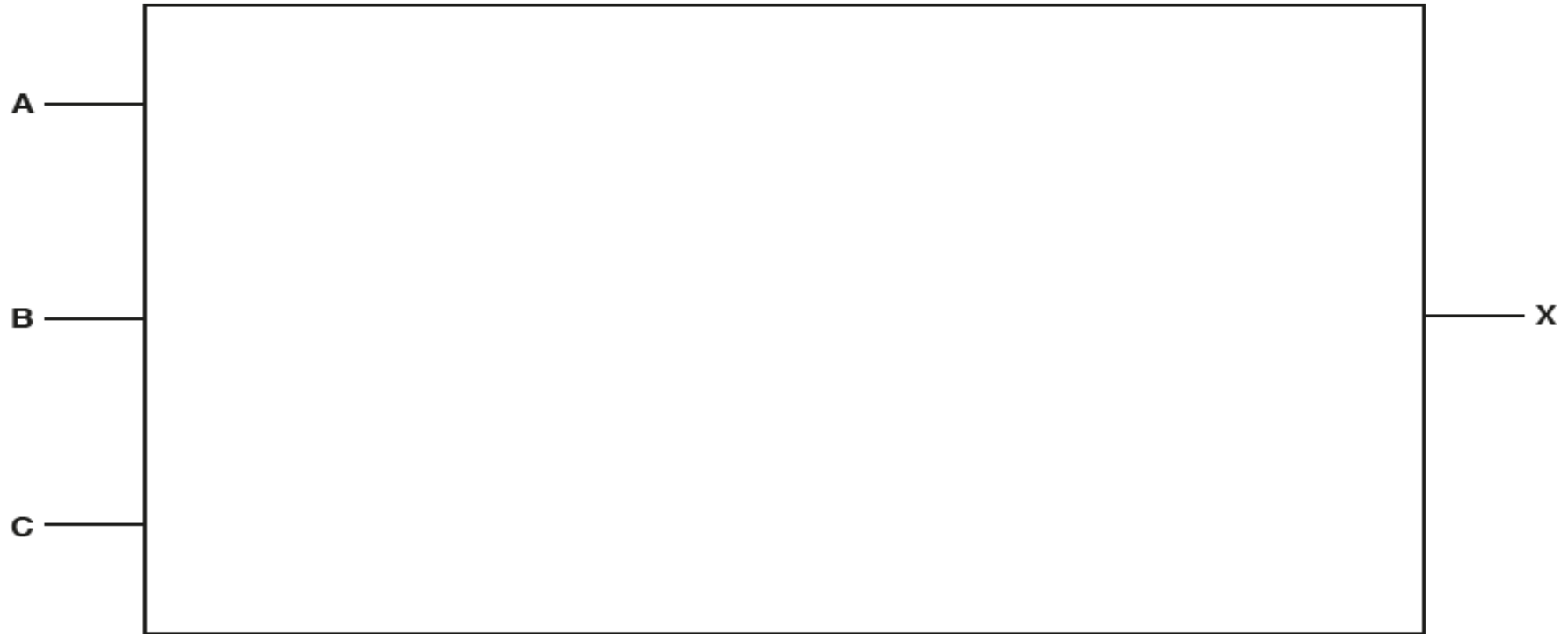
Question	Answer					Marks
6(b)	A	B	C	Working space	X	4
	0	0	0		1	
	0	0	1		0	
	0	1	0		1	
	0	1	1		1	
	1	0	0		1	
	1	0	1		1	
	1	1	0		1	
	1	1	1		0	

IGCSE Past Exam: Question 4

8 Consider the logic statement:

$X = 1$ if $((A \text{ is } 1 \text{ NOR } C \text{ is } 1) \text{ AND } (B \text{ is NOT } 1 \text{ NOR } C \text{ is } 1)) \text{ OR } (A \text{ is } 1 \text{ AND } B \text{ is } 1)$

- (a) Draw a logic circuit to match the given logic statement. Each logic gate used must have a maximum of **two** inputs. Do **not** attempt to simplify the logic statement.



[6]

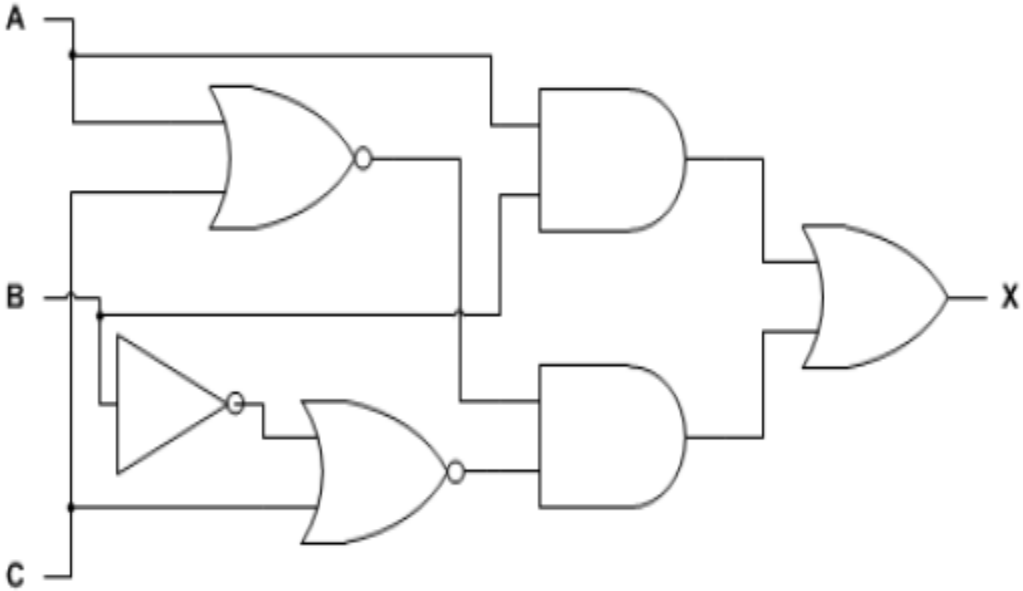
IGCSE Past Exam: Question 4

(b) Complete the truth table for the given logic statement.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

IGCSE Past Exam: Answer 4

Question	Answer	Marks
8(a)	1 mark per each correct logic gate, with correct input(s) 	6

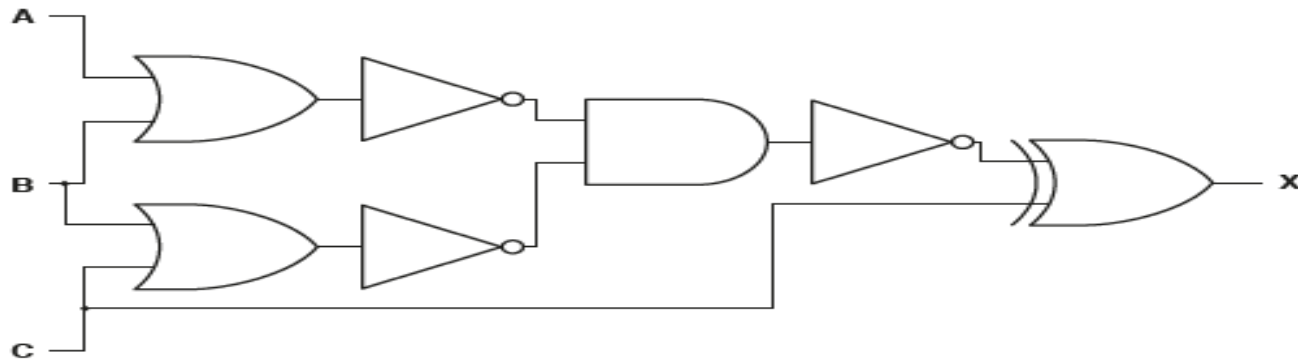
IGCSE Past Exam: Answer 4

PUBLISHED

Question	Answer	Marks																																													
8(b)	4 marks for 8 correct outputs 3 marks for 6/7 correct outputs 2 marks for 4/5 correct outputs 1 mark for 2/3 correct outputs <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	X	0	0	0		0	0	0	1		0	0	1	0		1	0	1	1		0	1	0	0		0	1	0	1		0	1	1	0		1	1	1	1		1	4
A	B	C	Working space	X																																											
0	0	0		0																																											
0	0	1		0																																											
0	1	0		1																																											
0	1	1		0																																											
1	0	0		0																																											
1	0	1		0																																											
1	1	0		1																																											
1	1	1		1																																											

IGCSE Past Exam: Question 5

4 Consider the given logic circuit:



(a) Redraw the logic circuit using only 4 logic gates. Each logic gate used must have a maximum of two inputs.



IGCSE Past Exam: Question 5

(b) Complete the truth table for the **given** logic circuit.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

(c) Describe the purpose of a logic gate in a logic circuit.

.....

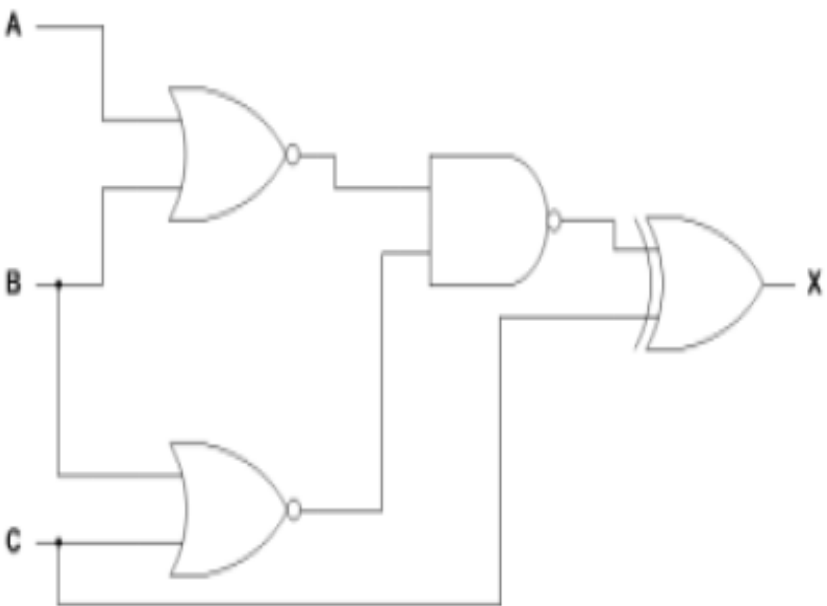
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[2]

IGCSE Past Exam: Answer 5

Question	Answer	Marks
4(a)	1 mark for each correct logic gate, with correct inputs:  <pre>graph LR; A --- OR1(()); B --- OR1; B --- OR2(()); C --- OR2; OR1 --- AND1(()); OR2 --- AND1; AND1 --- XOR1(()); XOR1 --- X[X];</pre>	4

IGCSE Past Exam: Answer 5

Question	Answer	Marks																																													
4(b)	4 marks for 8 correct outputs 3 marks for 6/7 correct outputs 2 marks for 4/5 correct outputs 1 mark for 2/3 correct outputs <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>	A	B	C	Working space	X	0	0	0		0	0	0	1		0	0	1	0		1	0	1	1		0	1	0	0		1	1	0	1		0	1	1	0		1	1	1	1		0	4
A	B	C	Working space	X																																											
0	0	0		0																																											
0	0	1		0																																											
0	1	0		1																																											
0	1	1		0																																											
1	0	0		1																																											
1	0	1		0																																											
1	1	0		1																																											
1	1	1		0																																											
4(c)	Two from: - To carry out a logical operation - To control the flow of electricity through a logic circuit - An input is given and the logic of the gate is applied to give an output // to alter the output from given inputs	2																																													

IGCSE Past Exam: Question 6

6 Consider the logic statement:

$X = 1$ if $((A \text{ is NOT } 1 \text{ OR } B \text{ is } 1) \text{ NOR } C \text{ is } 1) \text{ NAND } ((A \text{ is } 1 \text{ AND } C \text{ is } 1) \text{ NOR } B \text{ is } 1)$

(a) Draw a logic circuit to represent the given logic statement.



[6]

IGCSE Past Exam: Question 6

(b) Complete the truth table for the given logic statement.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

IGCSE Past Exam: Answer 6

Question	Answer	Marks
6(a)	1 mark for each correct logic gate (with the correct direction of input(s)) <pre>graph LR; A((A)) --- NOT1[NOT]; A --- OR1[OR]; B((B)) --- OR1; B --- AND1[AND]; B --- OR2[OR]; C((C)) --- AND1; NOT1 --- OR1; OR1 --- OR3[OR]; AND1 --- OR2; OR3 --- AND2[AND]; AND2 --- X((X))</pre>	6

IGCSE Past Exam: Answer 6

TABLE 1

Question	Answer	Marks																																													
6(b)	4 marks for 8 correct outputs 3 marks for 6 or 7 correct outputs 2 marks for 4 or 5 correct outputs 1 mark for 2 or 3 correct outputs <table><tr><td>A</td><td>B</td><td>C</td><td>Working space</td><td>X</td></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		1	0	1	1		1	1	0	0		0	1	0	1		1	1	1	0		1	1	1	1		1	4
A	B	C	Working space	X																																											
0	0	0		1																																											
0	0	1		1																																											
0	1	0		1																																											
0	1	1		1																																											
1	0	0		0																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		1																																											

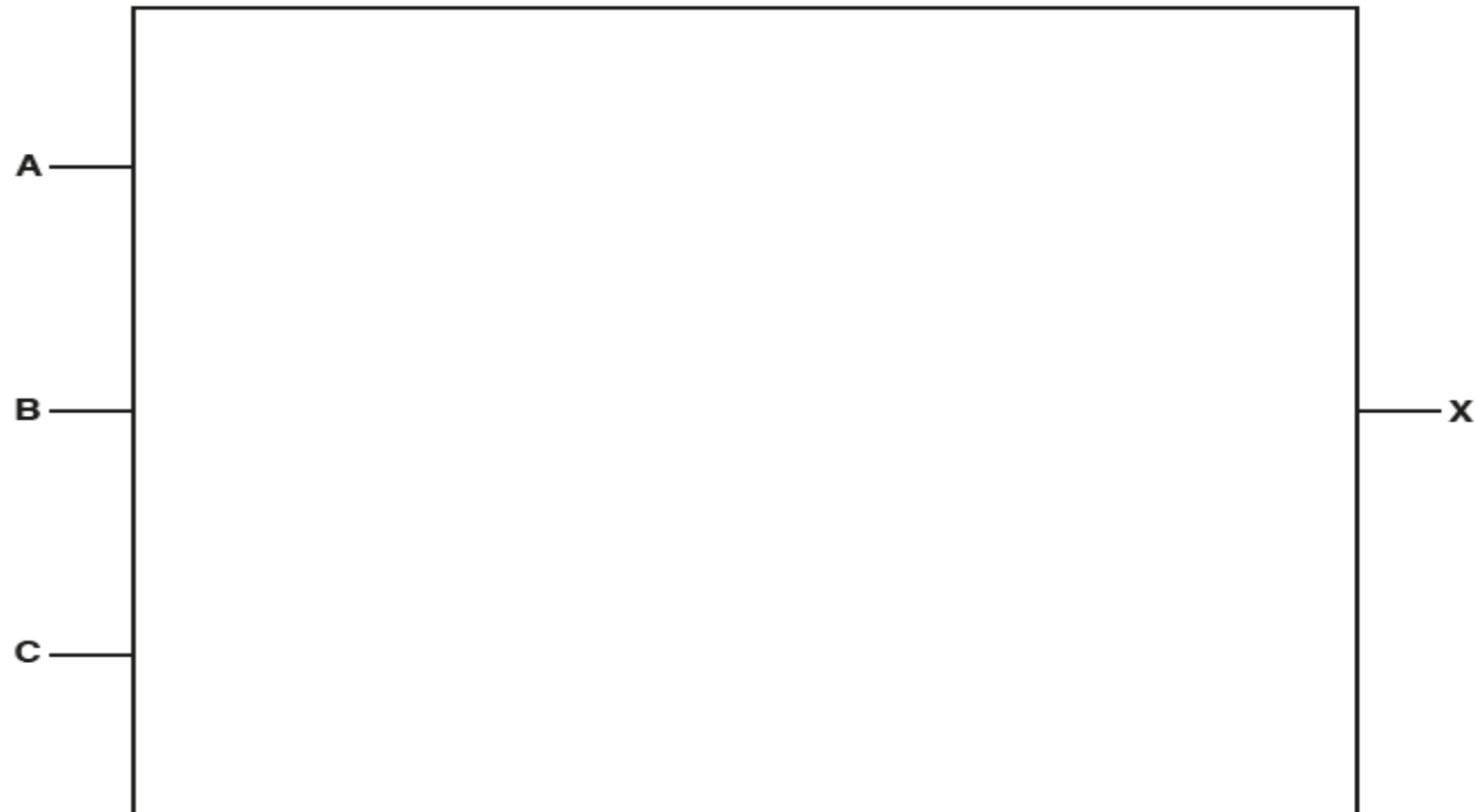
A	B	C	Working space	x
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

80

IGCSE Past Exam: Question 7

(b) Draw a logic circuit corresponding to the logic statement:

$X = 1$ if $((A \text{ is } 1 \text{ AND } B \text{ is } 1) \text{ AND } (A \text{ is } 1 \text{ OR } C \text{ is NOT } 1)) \text{ OR } (B \text{ is } 1 \text{ AND } C \text{ is NOT } 1)$



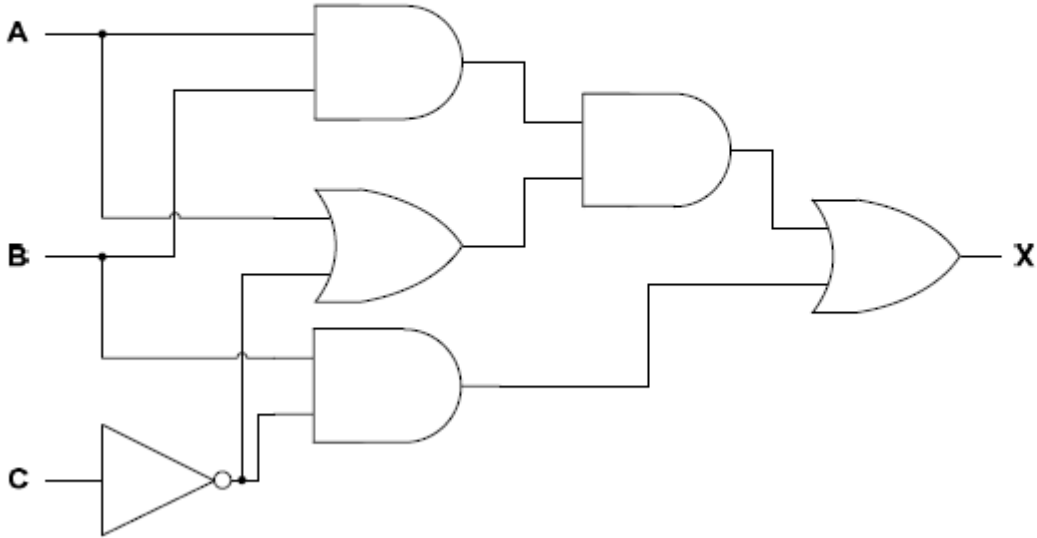
[6]

IGCSE Past Exam: Answer 7

PUBLISHED

Question	Answer	Marks																																				
10(a)	4 marks for 8 correct outputs 3 marks for 6 or 7 correct outputs 2 marks for 4 or 5 correct outputs 1 mark for 2 or 3 correct outputs <table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>	A	B	C	X	0	0	0	0	0	0	1	0	0	1	0	1	0	1	1	1	1	0	0	0	1	0	1	1	1	1	0	0	1	1	1	0	4
A	B	C	X																																			
0	0	0	0																																			
0	0	1	0																																			
0	1	0	1																																			
0	1	1	1																																			
1	0	0	0																																			
1	0	1	1																																			
1	1	0	0																																			
1	1	1	0																																			

IGCSE Past Exam: Answer 7

Question	Answer	Marks
10(b)	1 mark per correct gate with correct inputs. 	6

Past Paper Exam Question 8

(c) Computers consist of logic gates.

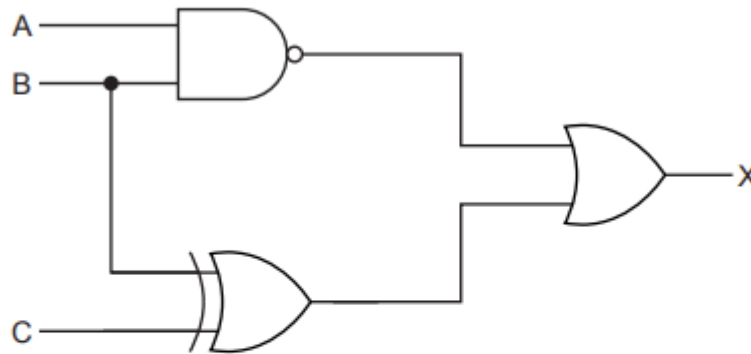
(i) Complete the table by writing **one** set of values (input 1 and input 2) for each gate that will give the output 1.

Gate	Input 1	Input 2	Output
AND			1
NAND			1
XOR			1
NOR			1

[4]

Past Paper Exam Question 8

(ii) Write the logic expression for the given logic circuit.



.....
.....
..... [3]

Past Paper Exam Answer 8

2(c)(i)	1 mark for each correct row. <table><tr><th>Gate</th><th>Input 1</th><th>Input 2</th></tr><tr><td>AND</td><td>1</td><td>1</td></tr><tr><td>NAND</td><td>0 0 1</td><td>0 1 0</td></tr><tr><td>XOR</td><td>0 1</td><td>1 0</td></tr><tr><td>NOR</td><td>0</td><td>0</td></tr></table>	Gate	Input 1	Input 2	AND	1	1	NAND	0 0 1	0 1 0	XOR	0 1	1 0	NOR	0	0	4
Gate	Input 1	Input 2															
AND	1	1															
NAND	0 0 1	0 1 0															
XOR	0 1	1 0															
NOR	0	0															
2(c)(ii)	1 mark per bullet point • A NAND B • B XOR C • OR (A NAND B) OR (B XOR C)	3															

Paper Exam Question 9

7 Complete the truth table for the following logic expression:

$$X = (A \text{ XOR } B) \text{ AND NOT } C$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

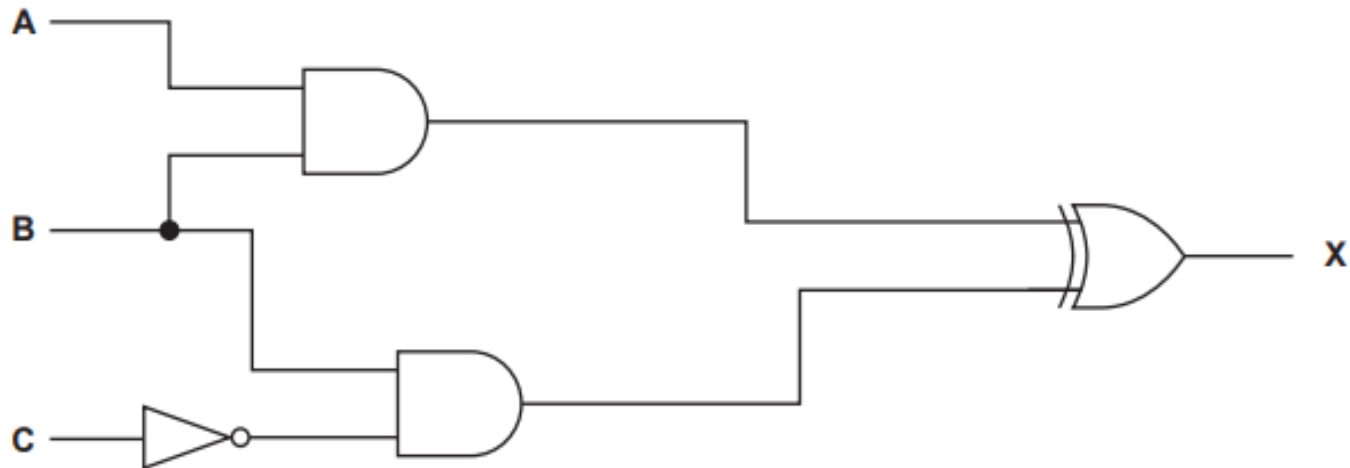
[2]

Past Paper Exam Answer 9

Question	Answer	Marks																																													
7	1 mark for first 4 rows, 1 mark for second 4 rows (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>	A	B	C	Working space	X	0	0	0		0	0	0	1		0	0	1	0		1	0	1	1		0	1	0	0		1	1	0	1		0	1	1	0		0	1	1	1		0	2
A	B	C	Working space	X																																											
0	0	0		0																																											
0	0	1		0																																											
0	1	0		1																																											
0	1	1		0																																											
1	0	0		1																																											
1	0	1		0																																											
1	1	0		0																																											
1	1	1		0																																											

Past Paper Exam Question 10

- 7 (a) Write the logic expression for the following logic circuit.



.....
..... [3]

Past Paper Exam Question 10

(b) Complete the truth table for the following logic expression:

$$X = (A \text{ NAND } B) \text{ OR } (A \text{ AND NOT } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

Past Paper Exam Answer 10

Question	Answer	Marks																																													
7(a)	1 mark for each section • A AND B • NOT C AND B // B AND NOT C • XOR (with remainder correct and bracketed and nothing extra) X = (A AND B) XOR (NOT C AND B)	3																																													
7(b)	1 mark for first 4 rows, 1 mark for second 4 rows (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		1	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		0	2
A	B	C	Working space	X																																											
0	0	0		1																																											
0	0	1		1																																											
0	1	0		1																																											
0	1	1		1																																											
1	0	0		1																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		0																																											

Past Paper Exam Question 11

8 Tick (✓) **one** box in each row to identify the logic gate that each statement describes.

Statement	AND	NAND	NOR	XOR	OR
The output is 1 only when both inputs are 1					
The output is 1 only when both inputs are different					
The output is 1 only when both inputs are 0					

[3]

Past Paper Exam Answer 11

Question	Answer	Marks
8	1 mark per correct row	3

Past Paper Exam Question 12

3 A logic expression is given:

$$S = (A \text{ AND } B \text{ AND } C) \text{ OR } (B \text{ XOR } C)$$

(a) Draw the logic circuit for the given expression.



Past Paper Exam Question 12

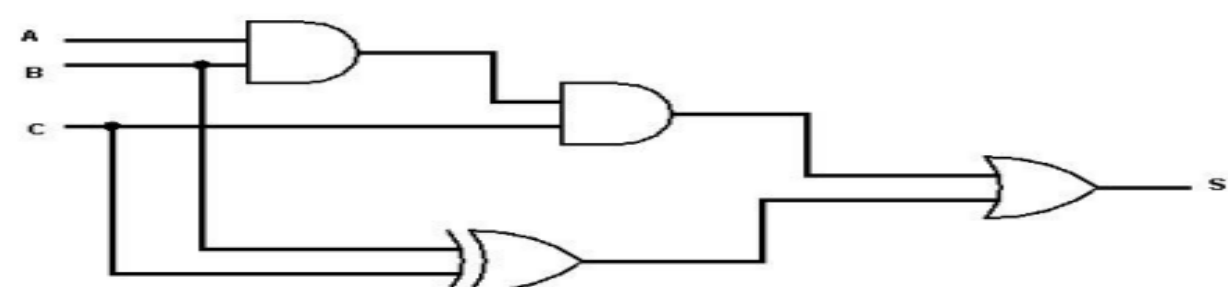
(b) Complete the truth table for the logic expression:

$$S = (A \text{ AND } B \text{ AND } C) \text{ OR } (B \text{ XOR } C)$$

A	B	C	Working space	S
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

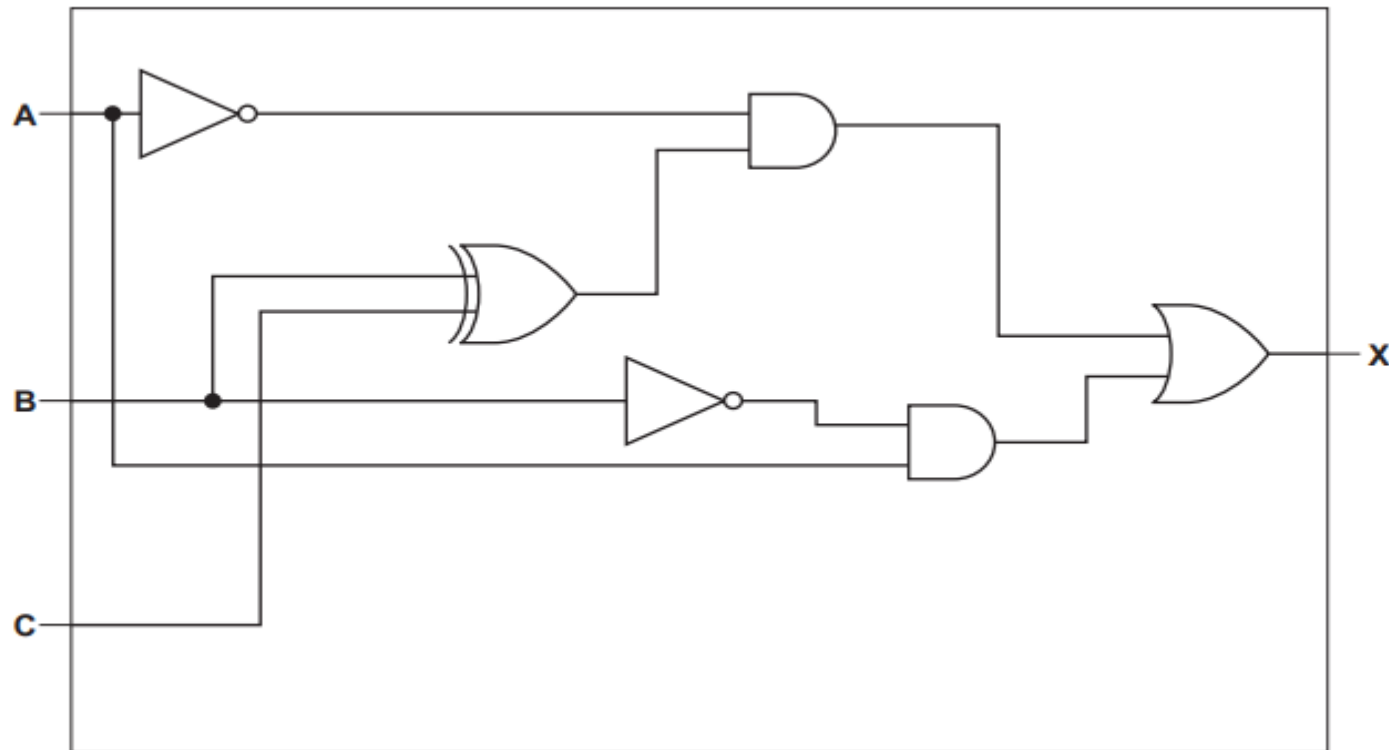
[2]

Past Paper Exam Answer 12

Question	Answer	Marks																																													
3(a)	1 mark for each correct gate, with correct inputs 	4																																													
3(b)	1 mark for each half (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>S</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	S	0	0	0		0	0	0	1		1	0	1	0		1	0	1	1		0	1	0	0		0	1	0	1		1	1	1	0		1	1	1	1		1	2
A	B	C	Working space	S																																											
0	0	0		0																																											
0	0	1		1																																											
0	1	0		1																																											
0	1	1		0																																											
1	0	0		0																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		1																																											

Past Paper Exam Question 13

5 (a) A logic circuit is given:



Past Paper Exam Question 13

Complete the following truth table for the logic circuit.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

Past Paper Exam Question 13

(b) Identify **one** logic gate not used in the logic circuit in **part (a)**.

Draw the symbol for this logic gate **and** complete its truth table.

Logic gate:

Symbol:

Truth table:

Input		Output
A	B	
0	0	
0	1	
1	0	
1	1	

[3]

Past Paper Exam Answer 13

Question	Answer	Marks																																													
5(a)	1 mark for each pair of correct answers (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>	A	B	C	Working space	X	0	0	0		0	0	0	1		1	0	1	0		1	0	1	1		0	1	0	0		1	1	0	1		1	1	1	0		0	1	1	1		0	4
A	B	C	Working space	X																																											
0	0	0		0																																											
0	0	1		1																																											
0	1	0		1																																											
0	1	1		0																																											
1	0	0		1																																											
1	0	1		1																																											
1	1	0		0																																											
1	1	1		0																																											

Past Paper Exam Answer 13

Question	Answer	Marks																																		
5(b)	1 mark for name, 1 mark for symbol, 1 mark for truth table - NAND  <table border="1"> <thead> <tr> <th colspan="2">Input</th><th rowspan="2">Output</th></tr> <tr> <th>A</th><th>B</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1</td><td>1</td></tr> <tr> <td>1</td><td>0</td><td>1</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </tbody> </table> - NOR  <table border="1"> <thead> <tr> <th colspan="2">Input</th><th rowspan="2">Output</th></tr> <tr> <th>A</th><th>B</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1</td><td>0</td></tr> <tr> <td>1</td><td>0</td><td>0</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </tbody> </table>	Input		Output	A	B	0	0	1	0	1	1	1	0	1	1	1	0	Input		Output	A	B	0	0	1	0	1	0	1	0	0	1	1	0	3
Input		Output																																		
A	B																																			
0	0	1																																		
0	1	1																																		
1	0	1																																		
1	1	0																																		
Input		Output																																		
A	B																																			
0	0	1																																		
0	1	0																																		
1	0	0																																		
1	1	0																																		

Past Paper Exam Question 14

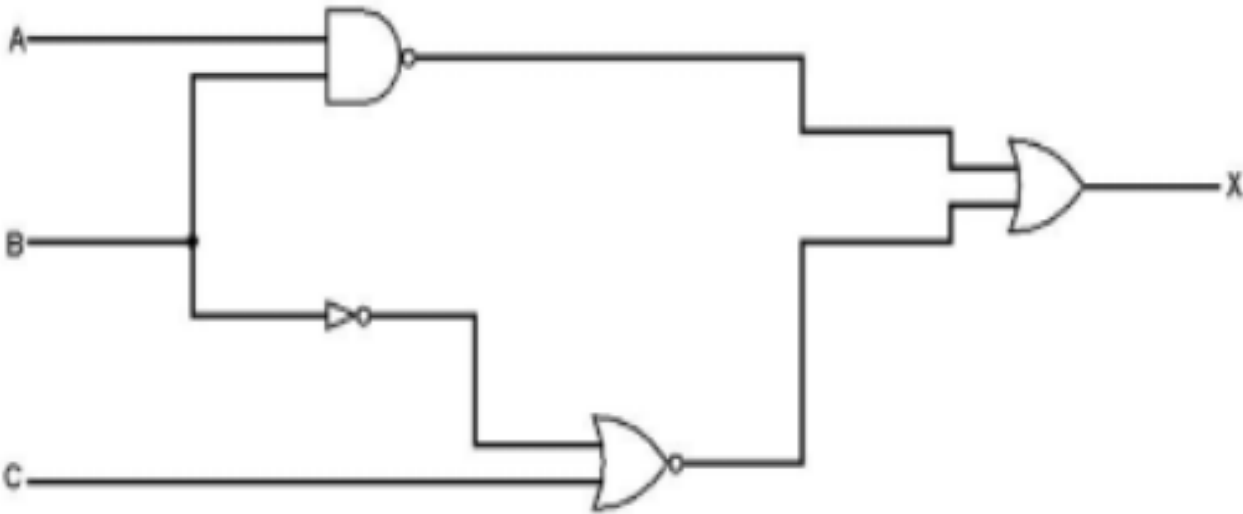
3 The following is a logic expression.

$$X = \text{NOT } (A \text{ AND } B) \text{ OR NOT } (\text{NOT } B \text{ OR } C)$$

Draw the logic circuit for the given expression using a maximum of **four** gates.



Past Paper Exam Answer 14

Question	Answer	Marks
3	1 mark for each correct gate $X = \text{NOT } (A \text{ AND } B) \text{ OR NOT } (\text{NOT } B \text{ OR } C)$ 	4

Past Paper Exam Question 14

- 3 (a) The following is a logic expression.

$$X = \text{NOT}(A \text{ OR } B) \text{ OR } (A \text{ AND } (B \text{ XOR } C))$$

Draw the logic circuit for the given expression, using a maximum of **four** logic gates.



Past Paper Exam Question 14

(b) Complete the truth table for the logic expression:

$$X = \text{NOT}(A \text{ OR } B) \text{ OR } (A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

Past Paper Exam Question 14

(c) The following is a logic expression.

A AND B XOR C OR NOT A

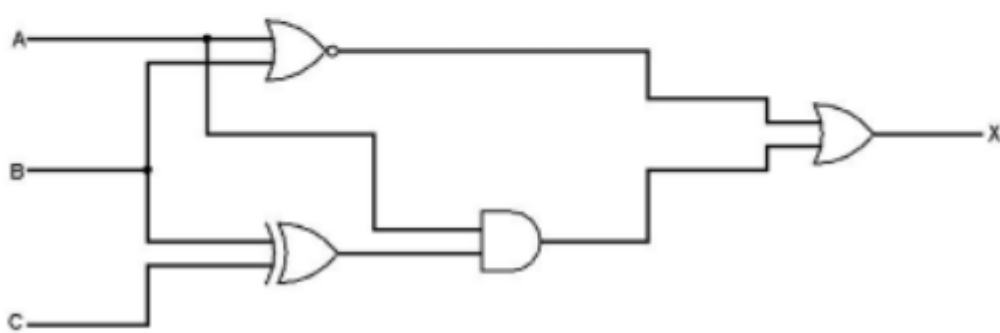
Identify **one** logic gate that would **not** be used in the logic circuit for this expression.
Draw the symbol for the logic gate.

Logic gate

Logic gate symbol:

[2]

Past Paper Exam Answer 14

Question	Answer	Marks
3(a)	1 mark per correct gate • A NOR B • B XOR C • A AND (B XOR C) • Final OR 	4

Past Paper Exam Answer 14

Question	Answer	Marks																																													
3(b)	1 mark for each correct pair of answers (4 shaded sections) <table><tr><th>A</th><th>B</th><th>C</th><th>Working Space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>	A	B	C	Working Space	X	0	0	0		1	0	0	1		1	0	1	0		0	0	1	1		0	1	0	0		0	1	0	1		1	1	1	0		1	1	1	1		0	4
A	B	C	Working Space	X																																											
0	0	0		1																																											
0	0	1		1																																											
0	1	0		0																																											
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Past Paper Exam Answer 14

3(c)	1 mark for name, 1 mark for symbol matching the name	2
	NAND 	
	NOR 	

Past Paper Exam Question 15

- 5 (a) Draw a logic circuit diagram for the logic expression:

$$X = \text{NOT } (A \text{ OR } B \text{ OR } C) \text{ OR } (B \text{ AND } C \text{ AND } D)$$



Past Paper Exam Question 15

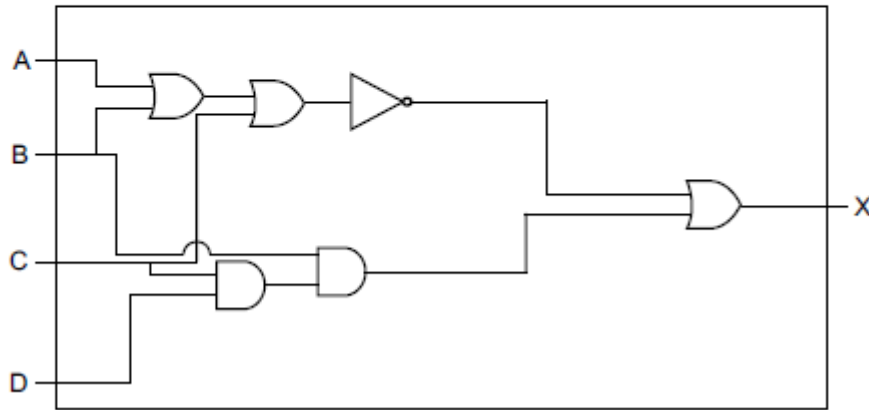
(b) Complete the truth table for the logic expression:

$$X = (A \text{ XOR } B) \text{ OR } \text{NOT } (A \text{ OR } B \text{ OR } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

Past Paper Exam Answer 15

Question	Answer	Marks	Guidance
5(a)	1 mark per bullet • A OR B OR C • NOT (A OR B OR C) • B AND C AND D • Final OR 	4	Accept working alternatives

Past Paper Exam Answer 15

Question	Answer	Marks																																													
5(b)	1 mark per pair of answers <table><tr><td>A</td><td>B</td><td>C</td><td>Working space</td><td>X</td></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		0	0	1	0		1	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		0	1	1	1		0	4
A	B	C	Working space	X																																											
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Past Paper Exam Question 16

- 4 (a) An alarm system (X) is enabled and disabled using either a switch (A) or a remote control (B). There are **two** infra-red sensors (C, D) and **one** door pressure sensor (E).

Parameter	Description of parameter	Binary value	Condition
A	Switch	1	Switch enabled
		0	Switch disabled
B	Remote control	1	Remote enabled
		0	Remote disabled
C	Infra-red sensor	1	Activated
		0	Not activated
D	Infra-red sensor	1	Activated
		0	Not activated
E	Door pressure sensor	1	Activated
		0	Not activated

The alarm sounds ($X = 1$) if the alarm is enabled **and** any one or more of the sensors is activated.

Past Paper Exam Question 16

Draw a logic circuit to represent the alarm system.



[3]

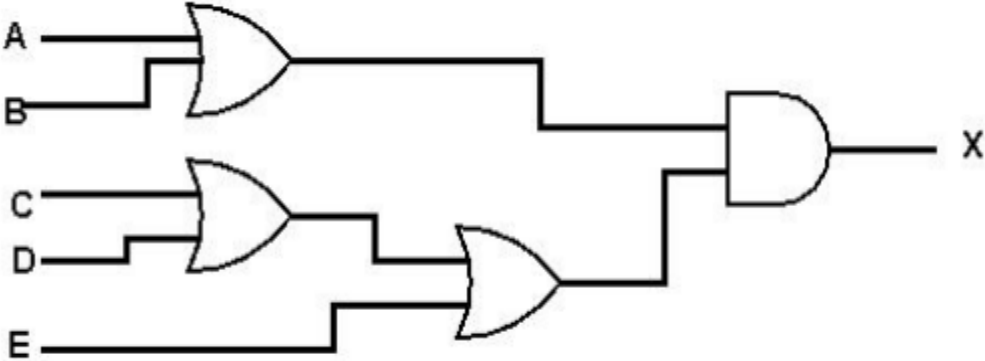
Past Paper Exam Question 16

(b) Complete the truth table for the logic expression: $X = A \text{ OR } (B \text{ XOR } C)$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

Past Paper Exam Answer 16

Question	Answer	Marks
4(a)	1 mark per bullet: • A OR B • C OR D OR E • Final AND  <pre> graph LR A --- OR1 B --- OR1 C --- OR2 D --- OR2 OR2 --- OR3 E --- OR3 OR1 --- AND OR3 --- AND AND --- X style OR1 fill:#fff,stroke:#000 style OR2 fill:#fff,stroke:#000 style OR3 fill:#fff,stroke:#000 style AND fill:#fff,stroke:#000 </pre>	3

Past Paper Exam Answer 16

4(b)

1 mark for each correct pair of rows

4

A	B	C	Working space	X
0	0	0		0
0	0	1		1
0	1	0		1
0	1	1		0
1	0	0		1
1	0	1		1
1	1	0		1
1	1	1		1

Past Paper Exam Question 17

6 (a) Draw a logic circuit to represent the logic expression:

$$X = A \text{ OR } (B \text{ AND NOT } C) \text{ OR } (A \text{ AND } B)$$



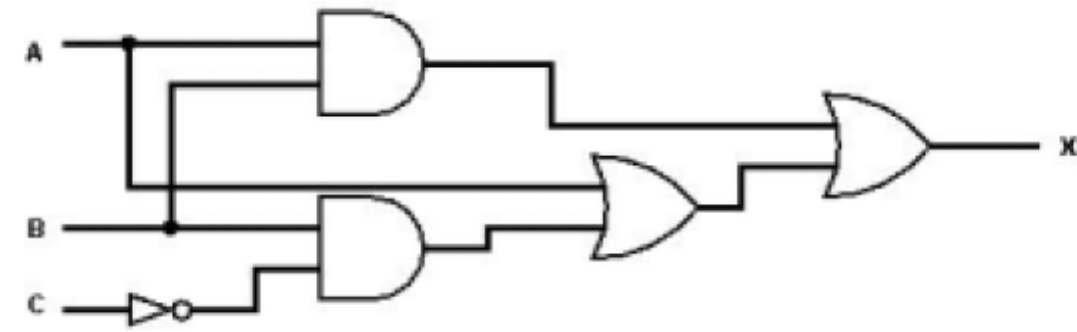
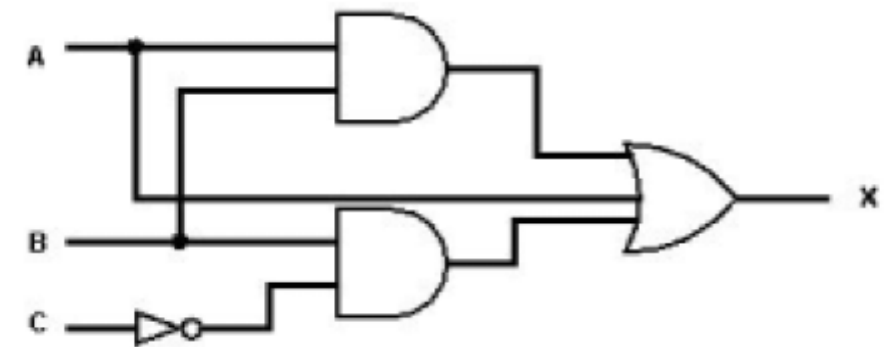
Past Paper Exam Question 17

(b) Complete the truth table for the logic expression in **part (a)**.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

Past Paper Exam Answer 17

Question	Answer	Marks
6(a)	1 mark per gate The OR gates may be re-sequenced  Alternatively: 3-input OR gate 1 mark for first three gates, and 2 marks for 3-input OR gate 	5

Past Paper Exam Answer 17

Question	Answer	Marks																																													
6(b)	1 mark for each pair of rows (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>Working Space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working Space	X	0	0	0		0	0	0	1		0	0	1	0		1	0	1	1		0	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		1	4
A	B	C	Working Space	X																																											
0	0	0		0																																											
0	0	1		0																																											
0	1	0		1																																											
0	1	1		0																																											
1	0	0		1																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		1																																											

Past Paper Exam Question 18

- 2 (a) A greenhouse control system has four input parameters (H, D, T, W) and two outputs (X, Y).

Parameter	Description of parameter	Binary value	Condition
H	Humidity	0	Too low
		1	Acceptable
D	Day	0	Night
		1	Day
T	Temperature	0	Too high
		1	Acceptable
W	Windows	0	Closed
		1	Open

The watering system turns on ($X = 1$) if:

either it is daytime **and** the temperature is too high

or the humidity is too low.

The fan turns on ($Y = 1$) if the temperature is too high **and** the windows are closed.

Past Paper Exam Question 18

Draw a logic circuit to represent the greenhouse control system.



[6]

Past Paper Exam Question 18

(b) Complete the truth table for the logic expression: $X = \text{NOT } A \text{ AND } (B \text{ NAND } C)$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

Past Paper Exam Answer 18

Question	Answer	Marks
2(a)	1 mark for each gate with the correct inputs. Final two gates must also have the correct output. <pre>graph LR; H --- NOT1[NOT]; NOT1 --- OR1[OR]; D --- AND1[AND]; T --- NOT2[NOT]; W --- NOT3[NOT]; NOT2 --- AND2[AND]; NOT3 --- AND2; AND1 --- AND3[AND]; OR1 --- AND3; OR1 --- X; AND3 --- Y</pre>	6

Past Paper Exam Answer 18

2(b)

One mark for each pair of rows.

4

A	B	C	Working space	X
0	0	0		1
0	0	1		1
0	1	0		1
0	1	1		0
1	0	0		0
1	0	1		0
1	1	0		0
1	1	1		0

Past Paper Exam Question 19

- 5 (a) A student needs to design a logic circuit to model the requirements for membership of a snooker club. Membership (X) depends on four criteria, as shown in the table:

Parameter	Description of parameter	Binary value	Condition
A	Over 18	1	True
		0	False
B	Recommended	1	True
		0	False
C	Full-time	1	True
		0	False
D	Retired	1	True
		0	False

Membership is approved ($X = 1$) if the person:

- is over the age of 18 **and** has been recommended by a pre-existing member **and**
- **either** is working full-time **or** is retired, but not both.

Paper Exam Question 19

Draw a logic circuit to represent the membership requirements.



[3]

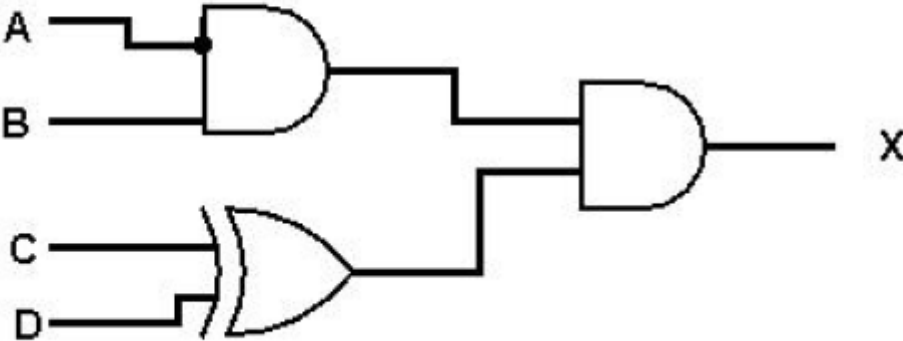
Past Paper Exam Question 19

(b) Complete the truth table for the logic expression: $X = (A \text{ XOR } B) \text{ AND NOT } C$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

Past Paper Exam Answer 19

Question	Answer	Marks
5(a)	1 mark per correct gate with correct inputs  <pre> graph LR A --- AND1[AND] B --- AND1 AND1 --- AND2[AND] C --- OR1[OR] D --- OR1 OR1 --- AND2 AND2 --- X </pre>	3

Past Paper Exam Answer 19

5(b)

1 mark for each correct pair of lines

4

A	B	C	Working space	X
0	0	0		0
0	0	1		0
0	1	0		1
0	1	1		0
1	0	0		1
1	0	1		0
1	1	0		0
1	1	1		0

Past Paper Exam Question 20

- 8 (a) Draw a logic circuit to represent the logic expression:

$$X = (A \text{ XOR } B) \text{ OR } (\text{NOT}(C \text{ AND } A))$$

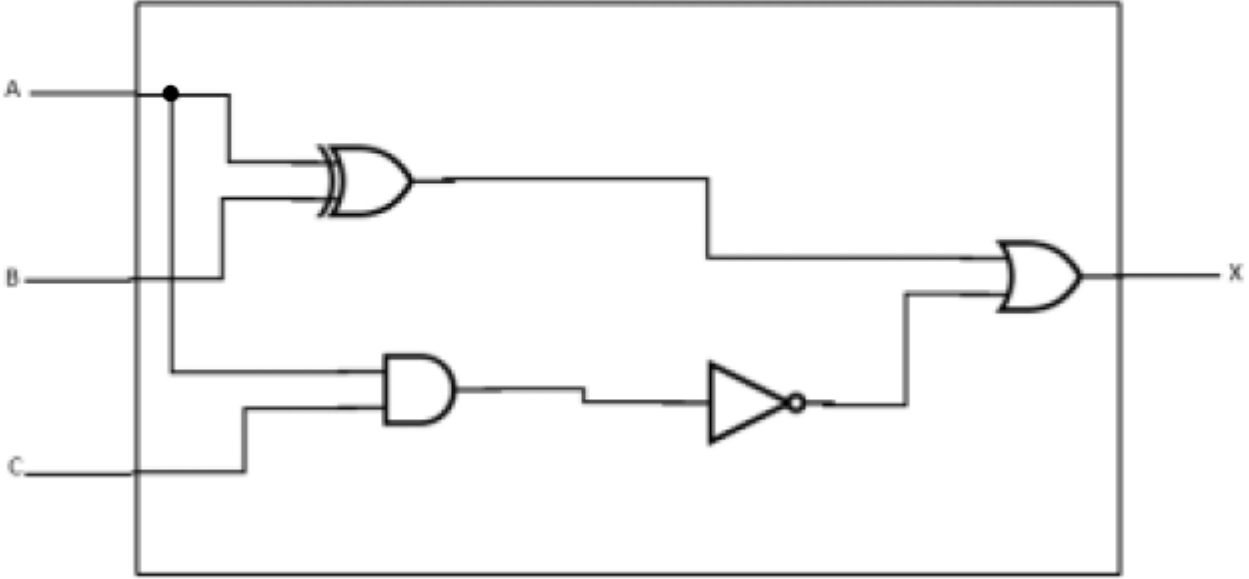


Past Paper Exam Question 20

(b) Complete the truth table for the logic expression in part (a).

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

Past Paper Exam Answer 20

Question	Answer	Marks
8(a)	1 mark for each logic gate with the correct inputs  <pre> graph LR A --- XOR1[] B --- XOR1 A --- AND1[] B --- AND1 C --- AND1 XOR1 --- OR1[] AND1 --- NOT1[] NOT1 --- OR1 OR1 --- X style XOR1 fill:none,stroke:none style AND1 fill:none,stroke:none style OR1 fill:none,stroke:none style NOT1 fill:none,stroke:none </pre>	4

Past Paper Exam Answer 20

Question	Answer	Marks																																													
8(b)	One mark for each correct pair of lines <table><tr><th>A</th><th>B</th><th>C</th><th>Working Space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>	A	B	C	Working Space	X	0	0	0		1	0	0	1		1	0	1	0		1	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		0	4
A	B	C	Working Space	X																																											
0	0	0		1																																											
0	0	1		1																																											
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1	0	1		1																																											
1	1	0		1																																											
1	1	1		0																																											

Past Paper Exam Question 21

5 A motor is controlled by a logic circuit. The circuit has inputs (0 or 1) from three sensors R, T and W. The motor is switched off when the output from the logic circuit is 1.

The following table shows the three sensors and the conditions being monitored.

Sensor	Description	Binary value	Condition
R	rotation	0	rotation < 4000 rpm
		1	rotation >= 4000 rpm
T	temperature	0	temperature >= 90 °C
		1	temperature < 90 °C
W	water flow rate	0	water flow rate >= 50 litre/min
		1	water flow rate < 50 litre/min

The output, X, is 1 if:

temperature >= 90 °C and rotation >= 4000 rpm

or

temperature < 90 °C and water flow rate >= 50 litre/min

Past Paper Exam Question 21

(i) Draw a corresponding logic circuit.



[5]

Past Paper Exam Question 21

(ii) Give a logic statement corresponding to the logic circuit in part (i).

.....
.....[2]

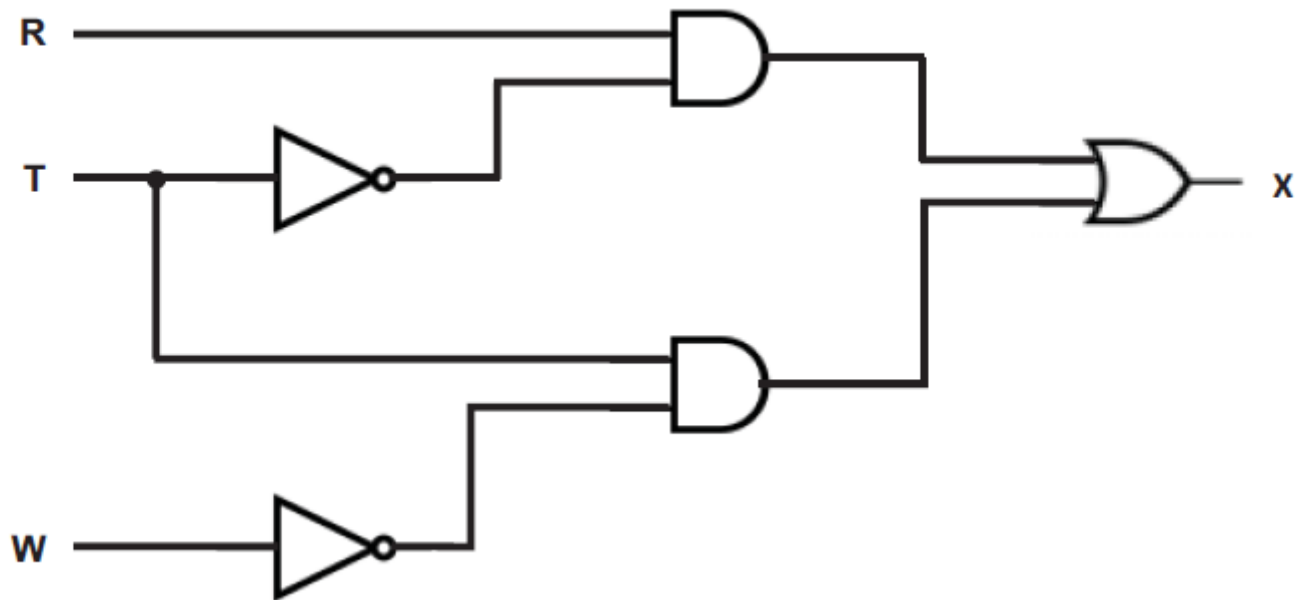
(iii) Complete the truth table for this system.

INPUT			Workspace	OUTPUT
R	T	W		X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

Past Paper Exam Answer 21

5 (a) (i) One mark for each correct gate.



[5]

Past Paper Exam Answer 21

(ii) $(R.\bar{T}) + (T.\bar{W})$ // (R AND NOT T) OR (T AND NOT W)

[2]

(iii) One mark for each pair of lines as shaded.

INPUT			Working space	OUTPUT X
R	T	W		
0	0	0		0
0	0	1		0
0	1	0		1
0	1	1		0
1	0	0		1
1	0	1		1
1	1	0		1
1	1	1		0

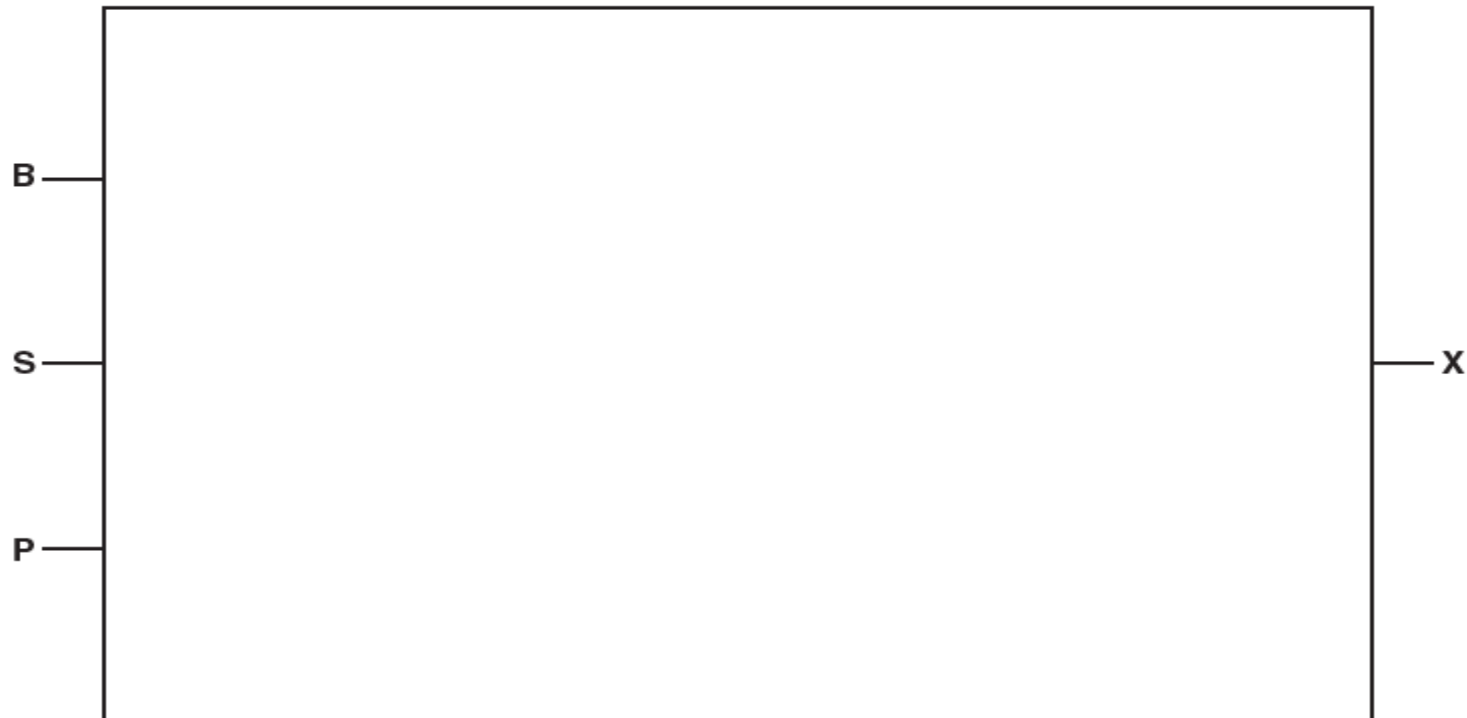
Past Paper Exam Question 22

- 1 (a) A student writes the following logic expression:

X is 1 IF (B is NOT 1 AND S is NOT 1) OR (P is NOT 1 AND S is 1)

Draw a logic circuit to represent this logic expression.

Do not attempt to simplify the logic expression.



Past Paper Exam Question 22

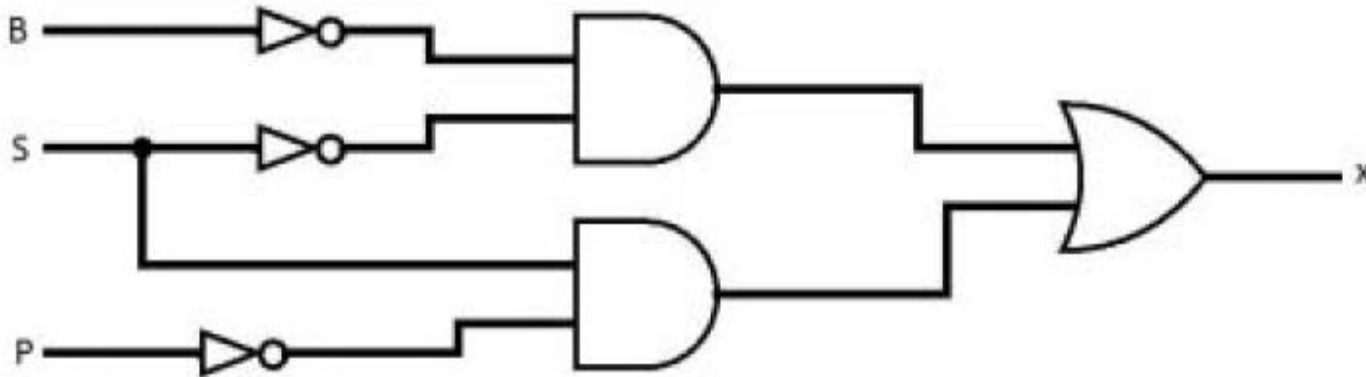
(b) Complete the truth table for the logic expression given in **part (a)**.

B	S	P	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

Past Paper Exam Answer 22

1 (a) **ONE** mark for each correct gate.



[6]

Past Paper Exam Answer 22

(b) **ONE** mark for each pair of rows.

B	S	P	Working space	X
0	0	0		1
0	0	1		1
0	1	0		1
0	1	1		0
1	0	0		0
1	0	1		0
1	1	0		1
1	1	1		0